	S.S.V.P.S's B. S. Deore Polytechnic, Dhule **Department of Computer Engineering** *Academic Year : 2025 – 2026*

Presentation on Capstone Project Execution (CPE–316004)

Topic	Traditional Attire Rental Shop Website	
Presented by	1. Kunal Raghunath Chaudhari.	
	2. Sakshi Vinod Chavan.	
	3. Harshali Vijay Fulpagare	
	4. Daksh Pravin Jadhav	
Guided by	Mr. N. D. Patel	
Semester	**Sixth (K Scheme)**	**Date :** 17/04/2026

Abstract

The *Traditional Attire Rental Shop Website* is a user-friendly platform designed to simplify the process of renting traditional clothing. It provides an affordable and convenient alternative to purchasing expensive outfits for one-time use. Users can easily browse, filter, and rent attire based on their size, preference, and occasion.

The system includes features like user registration, real-time availability tracking, booking management, and secure online payments. It ensures transparency in pricing by clearly displaying rent, deposits, and additional charges, while also reducing manual errors through digital record management.

The website is developed using modern technologies such as React.js, Vite.js, HTML, CSS, and Bootstrap for the frontend, Node.js and Express.js for the backend, MongoDB for database management, and Razorpay for secure payment integration.

Content

1. Introduction (background of the industry or used based problem / task)
2. Literature Survey
3. Scope of the project
4. Methodology
5. Details of designs, working and processes
6. Results and applications
7. Conclusion and future scope
8. References and Bibliography

1. Introduction

- **Traditional Attire Rental Shop**

The **Traditional Attire Rental Shop** started its business in **2019** and provides rental services for **Nauvari sarees, lehengas, traditional blouses, and jewellery such as necklaces, earrings, and accessories**.These products are mainly used during **weddings, festivals, haldi ceremonies, mehndi functions, receptions, and cultural programs**. Currently, the shop manages bookings and records using a **manual system such as registers and phone calls**, which becomes difficult during busy seasons.

Category	Examples
Nauvari Saree	• Cotton Nauvari Saree • Silk Nauvari Saree • Designer Nauvari Saree
Lehenga	• Bridal Lehenga • Designer Lehenga • Traditional Lehenga
Jewellery	• Necklace Sets • Earrings • Bangles
Other Collection	• Traditional Blouses • Waist Belt • Hair Accessories

2. Literature Review

The literature review focuses on traditional attire rental businesses such as *Namrata Nauvari Special Shop* located in Taloda, Nandurbar. It studies how existing systems operate, including manual processes and online rental platforms. These systems often rely on manual record keeping, which leads to errors, delays, and difficulty in managing bookings and customer details. The study highlights various limitations such as lack of real-time availability tracking, unclear pricing, and inefficient record management that affect efficiency and customer satisfaction.

The purpose of this study is to analyze real-world issues such as booking conflicts, payment handling problems, and lack of proper record management. It also considers problems like managing multiple bookings and handling peak season demand. This analysis helps in identifying the need for an improved and automated rental management system that provides accuracy, transparency, and better service.

2.1. Requirement Gathering

Questions Asked in Owner:

Q1. Do you face issues like double booking due to manual record updates?

Answer: Yes, sometimes double booking happens because records are maintained manually and not updated properly.

Q2. Do you experience delays in the booking confirmation?

Answer: Yes, booking confirmation takes time because availability needs checked manually.

Q3. Is there any confusion in pricing, rent, deposit, or extra charges?

Answer: Yes, sometimes customers get confused about total charges as everything is not clearly calculated.

Q4. Do you face problems in managing payments or tracking transactions?

Answer: Yes, it is difficult to track advance payments, pending amounts, and final payments manually.

2.1. Requirement Gathering

Q5. Is it difficult to track booking history and previous records?

Answer: Yes, searching old records in registers takes time.

Q6. Do you have a proper complaint handling or customer support system?

Answer: No, there is no proper complaint system, issues handled manually.

Q7. Do you find it difficult to check product availability in real-time?

Answer: Yes, availability is checked manually, so it is not always accurate.

Q8. Is managing customer details manually time-consuming?

Answer: Yes, writing and maintaining customer details manually takes a lot of time.

Q9. Do you face issues while handling returns and inspection records?

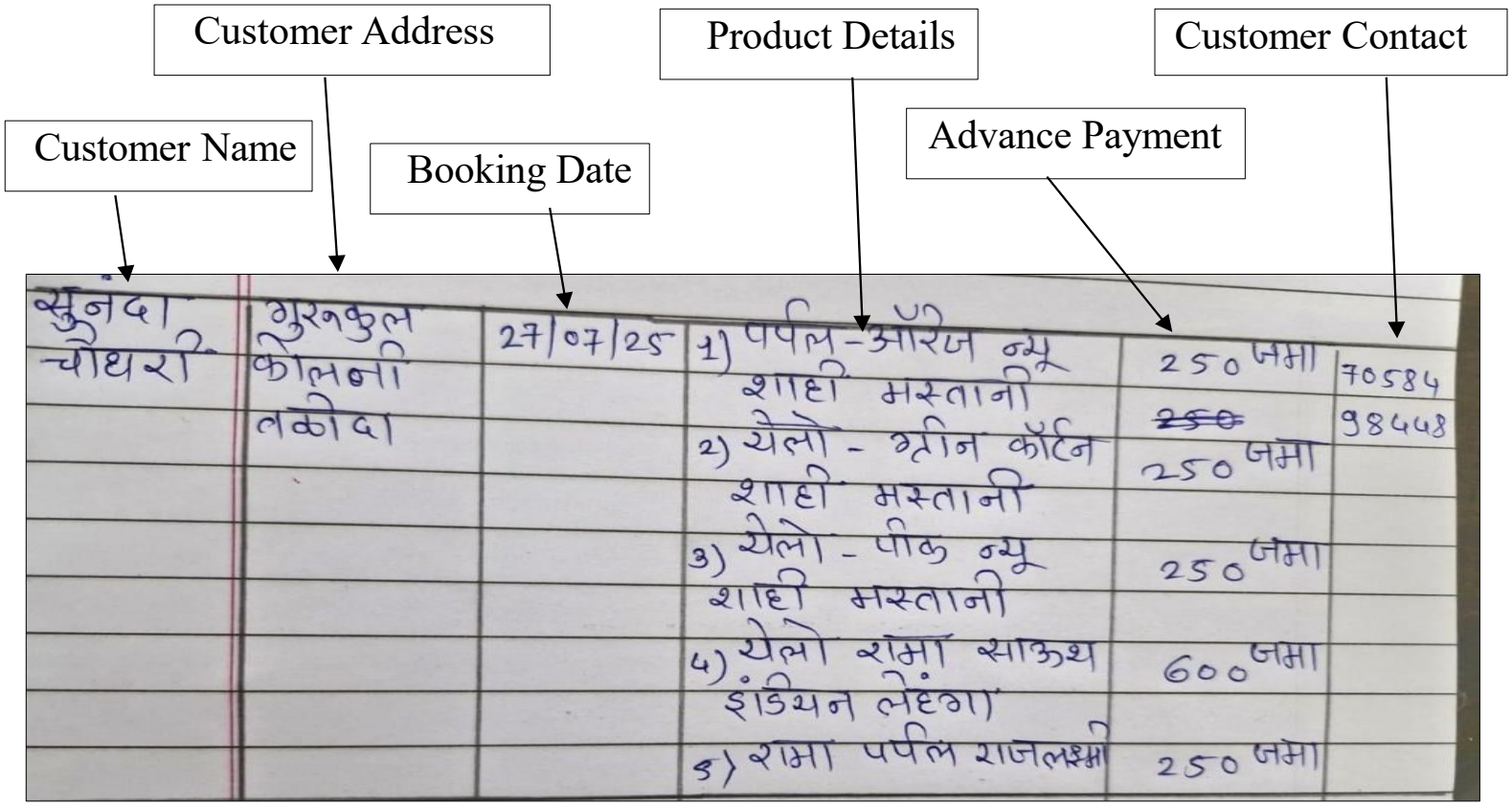
Answer: Yes, inspection details are not recorded properly, which creates confusion after return.

2.2 Study of Existing System

This section focuses on analyzing the current rental systems used in traditional attire businesses. These systems include manual methods as well as partially digital platforms. Most shops rely on manual record keeping for managing bookings, customer details, payments, and returns.

The study highlights that existing systems lack automation, real-time availability tracking, and proper data management. This leads to issues such as errors in booking, difficulty in maintaining records, delays in service, and reduced efficiency. It shows the need for a more organized and digital system to improve overall performance and user experience.

2.2.1 Manual Process System



सुनंदा चौधरी	गुरुकुल कोलनी वकोदा	27/07/25	1) पर्पल-ऑरेंज न्यू शाही मस्तानी	250 जमा	70584
			2) येलो - ग्रीन कॉटन शाही मस्तानी	~~250~~ 250 जमा	98448
			3) येलो - पीक न्यू शाही मस्तानी	250 जमा	
			4) येलो रामा साऊथ इंडियन लेहंगा	600 जमा	
			5) रामा पर्पल राजलक्ष्मी	250 जमा	

Figure 2.1 Manual Booking Record

2.2.2 Product Categories Structure

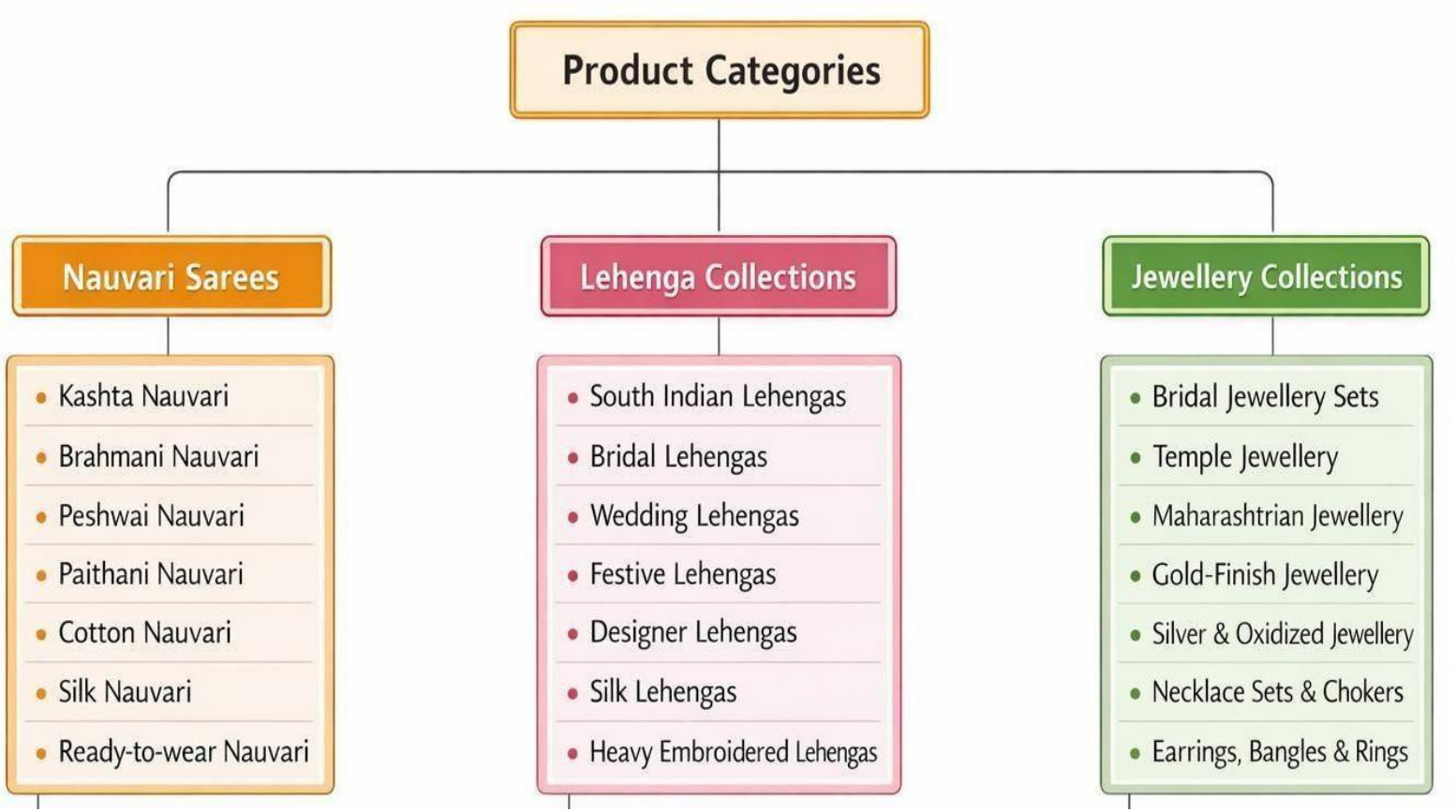


Figure 2.2 Product Categories Structure

2. 3 Limitation of Existing System

Sr. No.	Area	Limitations
1	Availability	Double booking due to manual updates.
2	Booking Approval	Delay in confirmation.
3	Pricing Transparency	Lack of clarity in rent + deposit + extra charges breakdown.
4	Payment Handling	Refund issues, payment failures, and unclear transaction records.
5	Record Keeping	Poor tracking of booking history and absence of payment proofs.
6	User Trust	Fake or poor quality products.
7	Customer Support	No complaint system.

Table 2.1: Limitations of Existing Rental Systems

2.4 Problem Identification

Stakeholder	Problem Identified
User Problems	Difficulty in finding suitable rentals quickly due to weak search filtering.
	Confusion in pricing due to hidden charges or unclear deposit/refund rules.
	Booking failures caused by availability mismatch (double booking).
Owner/Admin Problems	Manual handling of booking requests leads to delays and errors.
	Difficulty in managing cancellations, refunds, and disputes.
	Lack of a complete digital record system increases risk of disputes.

Table 2.2: Problem Identification for Users and Owner/Admin

2.5 Need of a System

Sr. No.	Role	Need of System
1	User	Easy search & filter (location, price, category)
2	User	Real-time availability tracking
3	User	Transparent pricing (rent + deposit + charges)
4	Owner/Admin	Booking & listing management dashboard
5	Owner/Admin	Digital records of bookings & payments
6	Owner/Admin	Verification & dispute handling system

Table 2.3: Need of System

2.5 Problem Definition

Managing a rental business through a web application is challenging because the system must track listings, availability, bookings, and payments accurately. Problems like double booking, delayed approvals, and unclear records can reduce efficiency and cause confusion. Therefore, a Rental Web Application is needed to provide real-time availability, proper booking tracking, and digital records, making the system simple, accurate, and easy to manage.

3. Scope of the project

Sr. No.	Functionality	Admin	Staff	Customer
1	Login and authentication	Yes	Yes	Yes
2	View product catalog	Yes	Yes	Yes
3	Add or update categories	Yes	No	No
4	Add or update products	Yes	Limited	No
5	Manage customer records	Yes	Yes	Own profile
6	Create booking	Yes	Yes	Yes
7	Confirm / update booking status	Yes	Yes	No
8	Check product availability	Yes	Yes	Yes
9	Pre-issue inspection entry	Yes	Yes	No
10	Return & post-return inspection	Yes	Yes	No
11	Payment entry and settlement	Yes	Yes	No
12	View my bookings	No	No	Yes

Table 3.1 System Functionalities According to Users

3.1 Functional Scope

Figure 3.1 Functional Scope

3.2 In-Scope and Out-of-Scope

Sr. No.	Area	In Scope	Out of Scope
1	Product Handling	Product listing, availability tracking, category mapping	IoT-based physical tracking devices
2	Booking	Booking creation, update, cancellation, status tracking	AI-based demand prediction
3	Returns	Return entry, inspection, damage & delay recording	Automated hardware-based inspection
4	Payments	Basic payment records, due tracking, settlement	Full accounting ERP integration
5	Users	Role-based login and user management	Biometric attendance or HR modules
6	Reports	Operational and financial summary reports	Advanced BI and forecasting dashboards
7	Customer Side	Browse products, create & view bookings	Marketplace-style multi-vendor platform

Table 3.2 Scope Boundary (In and out scope)

3. 3 User Requirements

Functionality	Admin	Staff	Customer
User Management	Add, update, remove users, assign roles	No access	Manage own profile
Product Management	Manage categories & products	Limited access	View products
Booking Management	Monitor all bookings	Create & update bookings	Create bookings
Availability Check	Yes	Yes	Yes
Inspection & Returns	Monitor process	Handle inspection & returns	No access
Payment Management	Verify payments & settlements	Record payments	View payment details
Reports	Access all reports	View basic reports	No access
Communication	System control	Operational updates	Receive notifications

Table 3.3: User Requirements Comparison (Admin vs Staff vs Customer)

3.4 Non-Functional Scope

The system also includes non-functional requirements to ensure quality, performance, and usability.

Performance: Fast response time for searching, booking, and payment processing

Reliability: Consistent and accurate data across all modules

Security: Role-based access control and secure authentication

Usability: Simple and user-friendly interface for all users

Scalability: Ability to handle growth in users, products, and bookings

Maintainability: Modular design for easy updates and enhancements

4. Methodology

This section explains the methodology used to develop the Traditional Attire Rental Shop Management System. The project follows the **Waterfall Model of SDLC**, where development is carried out step by step, including requirement analysis, design, implementation, testing, deployment, and maintenance.

Requirement analysis was done through observation and discussions to identify issues like booking conflicts, poor record management, and payment tracking problems. Based on these findings, the system was designed with modules such as customer management, booking, payment, return handling, and inventory tracking.

The system is developed using modern technologies like React.js and Vite.js for frontend, Node.js and Express.js for backend, MongoDB for database, and Razorpay for payment integration, ensuring efficiency, accuracy, and improved user experience.

4.1 Software Development Methodology

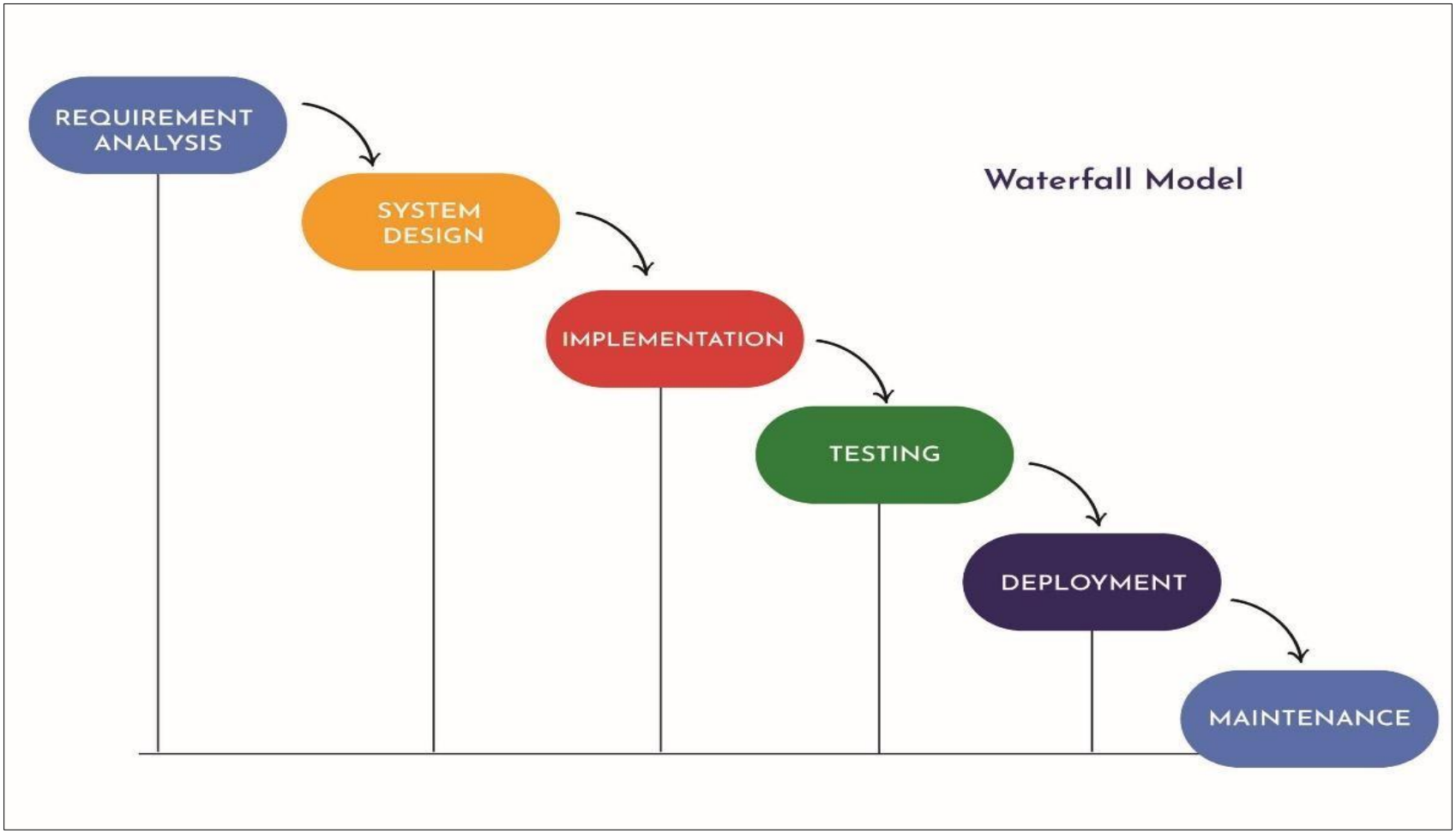


Figure 4.1: SDLC Waterfall Model — Sequential Phase Flow

4.2 User Requirement Analysis

Functionality	Admin	Staff	Customer
User Management	Manage users and roles	No access	Manage own profile
Product Management	Manage categories and products	Limited access	View products
Booking Management	Monitor all bookings	Create & update bookings	Create bookings
Availability Check	Yes	Yes	Yes
Inspection & Returns	Monitor process	Handle inspection & returns	No access
Payment Management	Verify payments	Record payments	View payment details
Reports	Access all reports	View basic reports	No access

Table 4.1 User Requirements Analysis Table

4.3 System Design Methodology

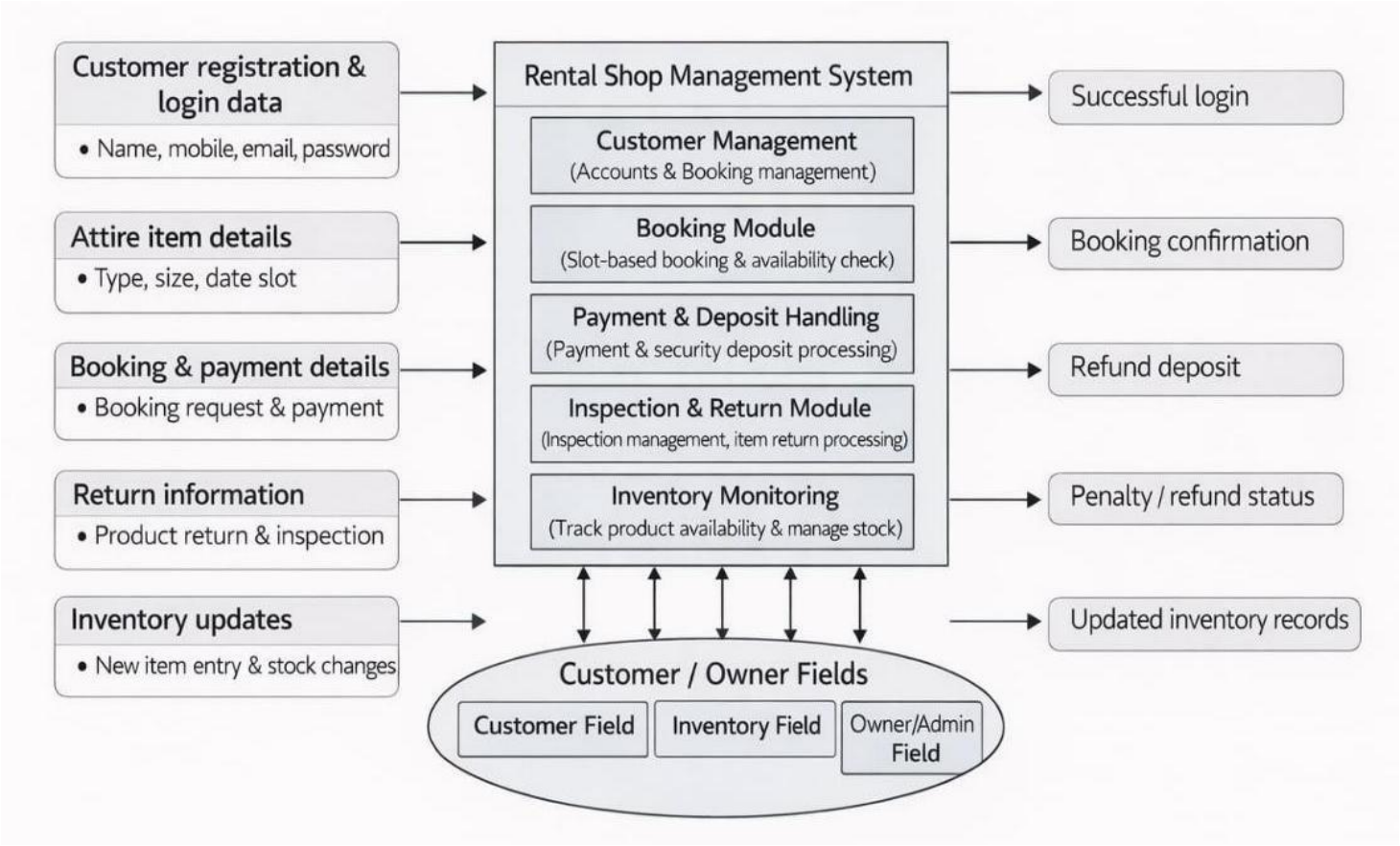


Figure 4.2: Proposed System for Traditional Attire Rental Shop Management System

4.4 Technology use to develop project

- **Programming Language(s):** JavaScript
- **Frontend:** HTML, CSS, Bootstrap, React.js (using Vite.js)
- **Backend:** Node.js, Express.js
- **Database:** MongoDB
- **Payment Integration:** Razorpay (for online payment processing)
- **Development Tools:** GitHub (Version Control), Postman (API Testing), Visual Studio Code (Development)
- **Operating System:** Windows 11

5. Details of Designs, Working and Processes

This section explains how the Rental Management System is designed and how it works internally. It describes the interaction between different modules such as Admin, Staff, and Customer, and how data flows through the system.

The system manages the complete rental process from product setup to booking, return, payment, and report generation. Each module performs specific tasks, while the database stores and updates all information in real time.

Overall, the system ensures smooth operations, reduces manual errors, and provides accurate and efficient rental management.

5.1 System Flow Diagram

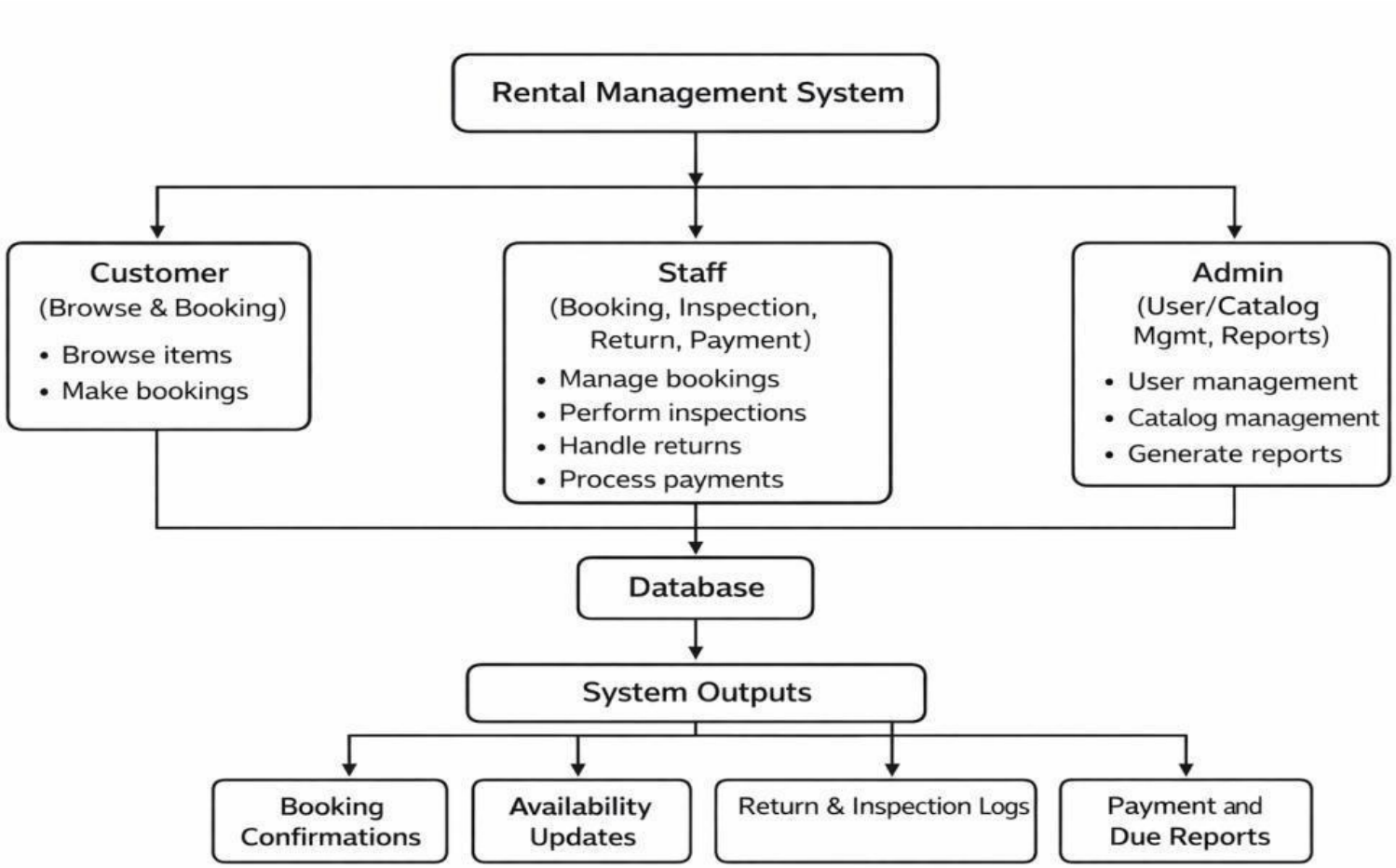


Figure 5.1: System Flow Diagram — Actors, System Core, and Outputs

5.2 Admin Module

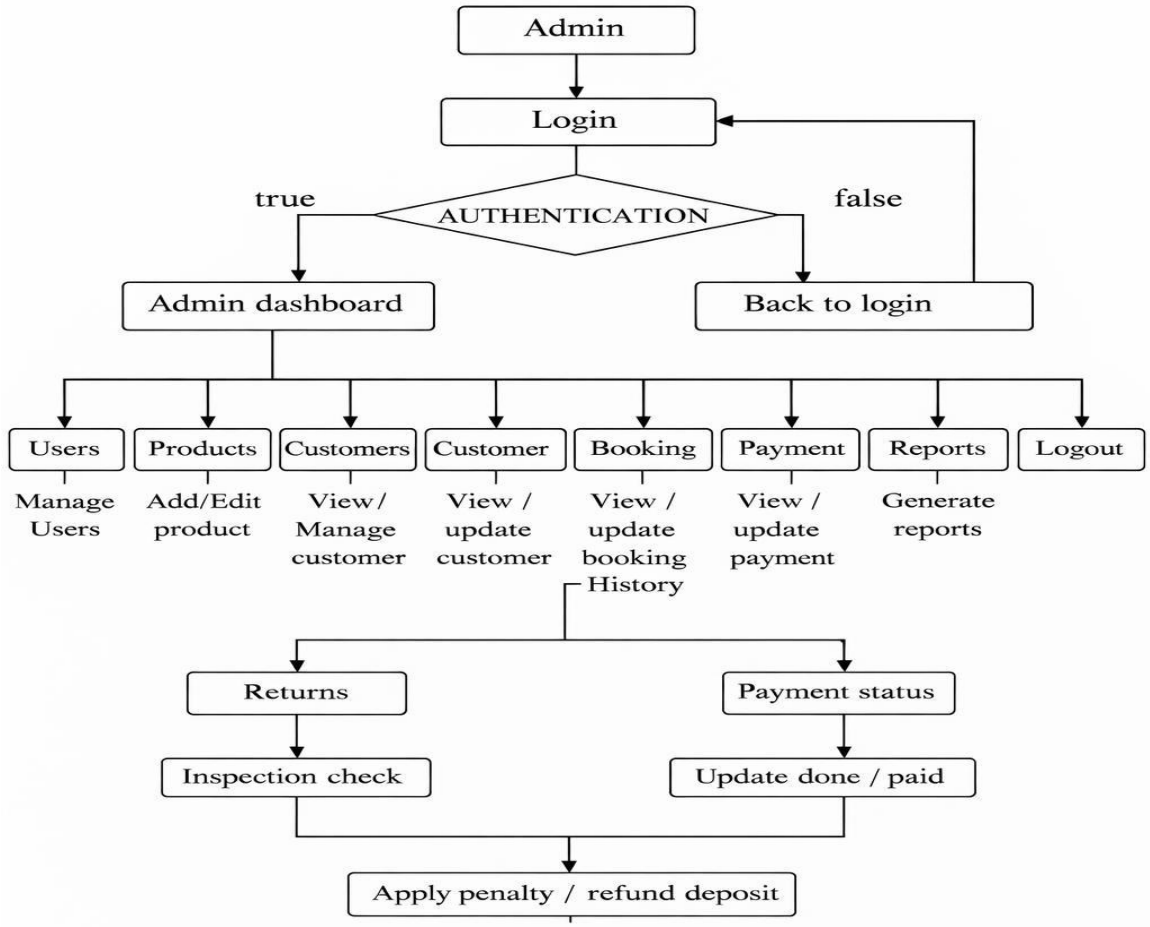


Figure 5.2: Admin Module Flow — Login to Report Generation and Logout

5.3 Staff Module

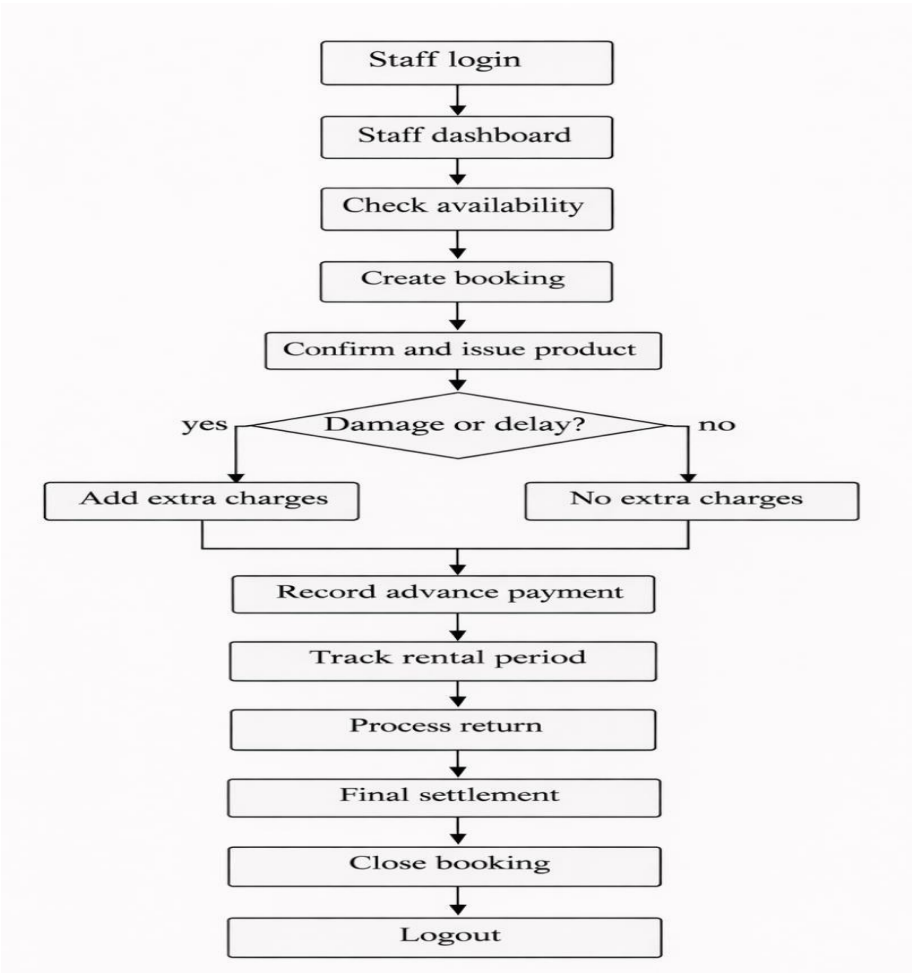


Figure 5.3 : Staff Module Flow

5.4 Customer Module

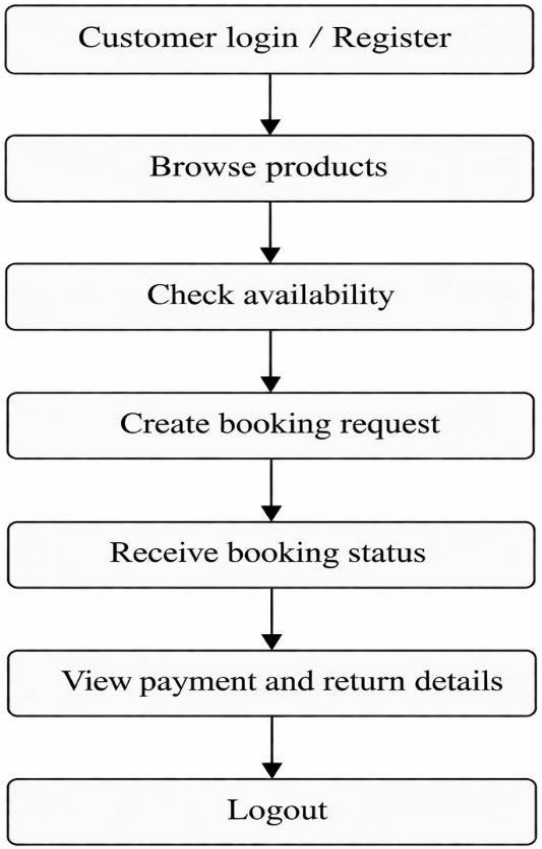


Figure 5.4 : Customer Module Flow

5.5 Use case Diagram

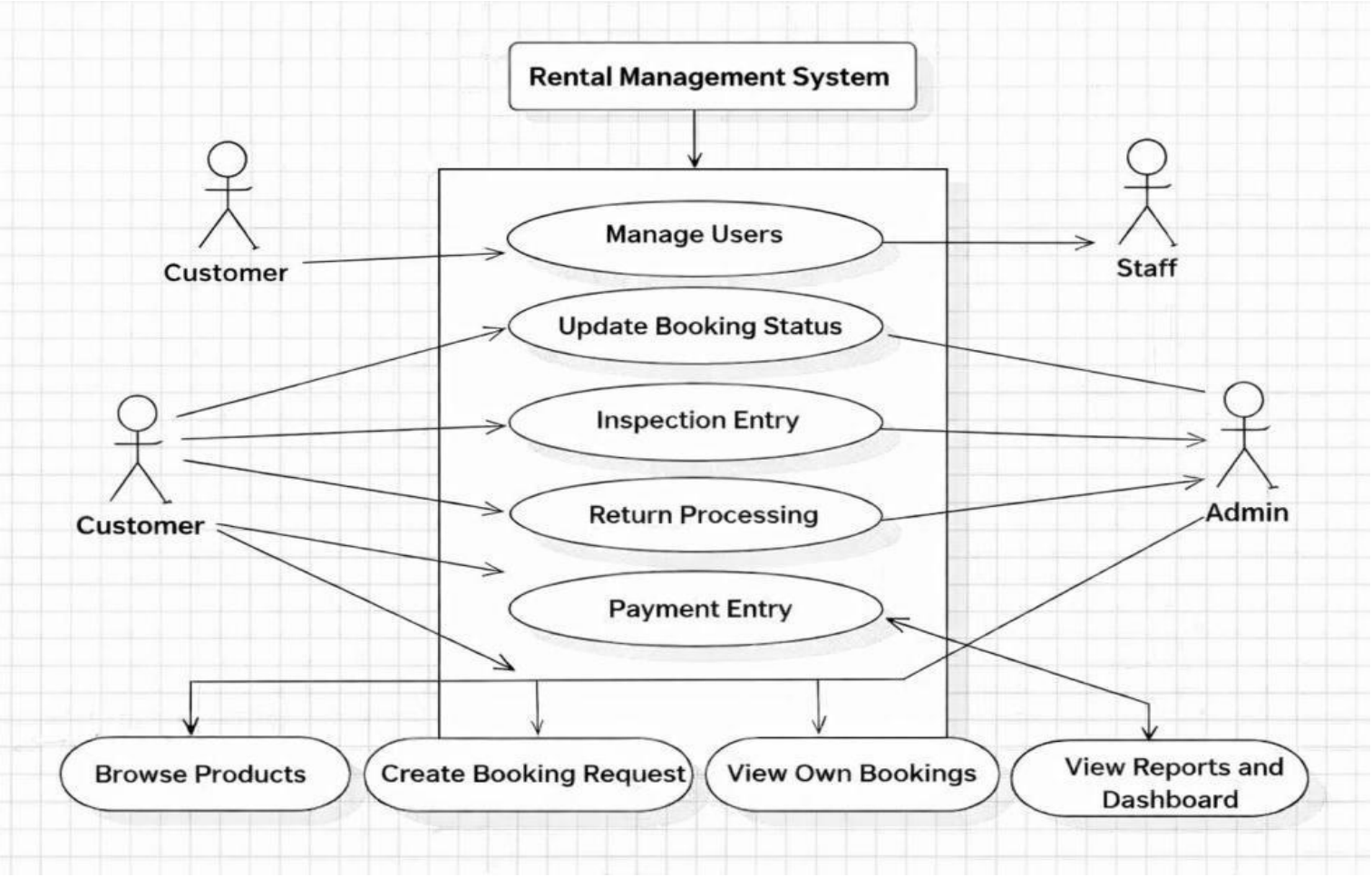


Figure 5.5: Use Case Diagram — System Actors and Role-wise Use Cases

5.6 Database Schema

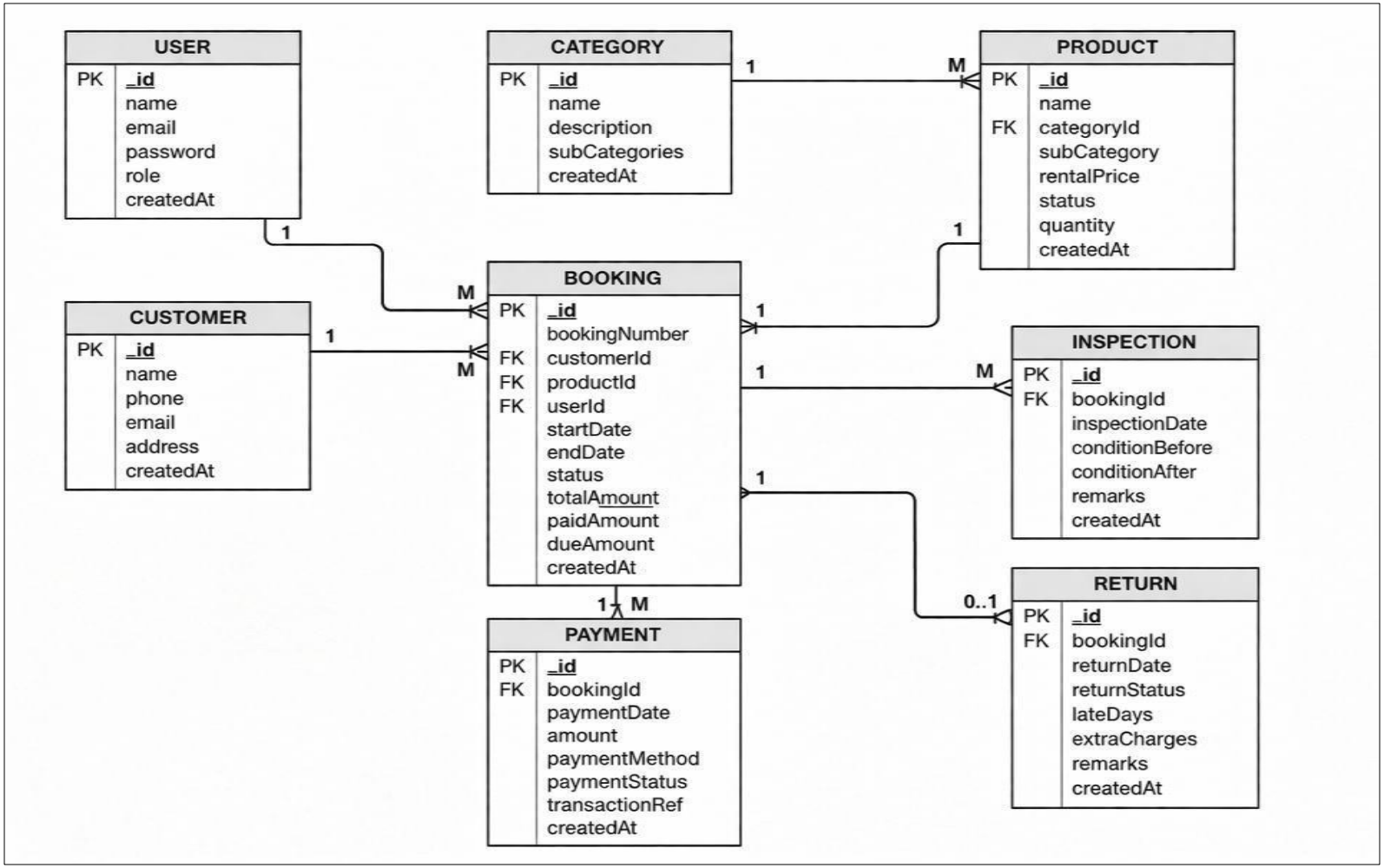


Figure 5.6 : Database ER Diagram — Entity Descriptions and Key Fields

5.7 RazorPay Integration Flow

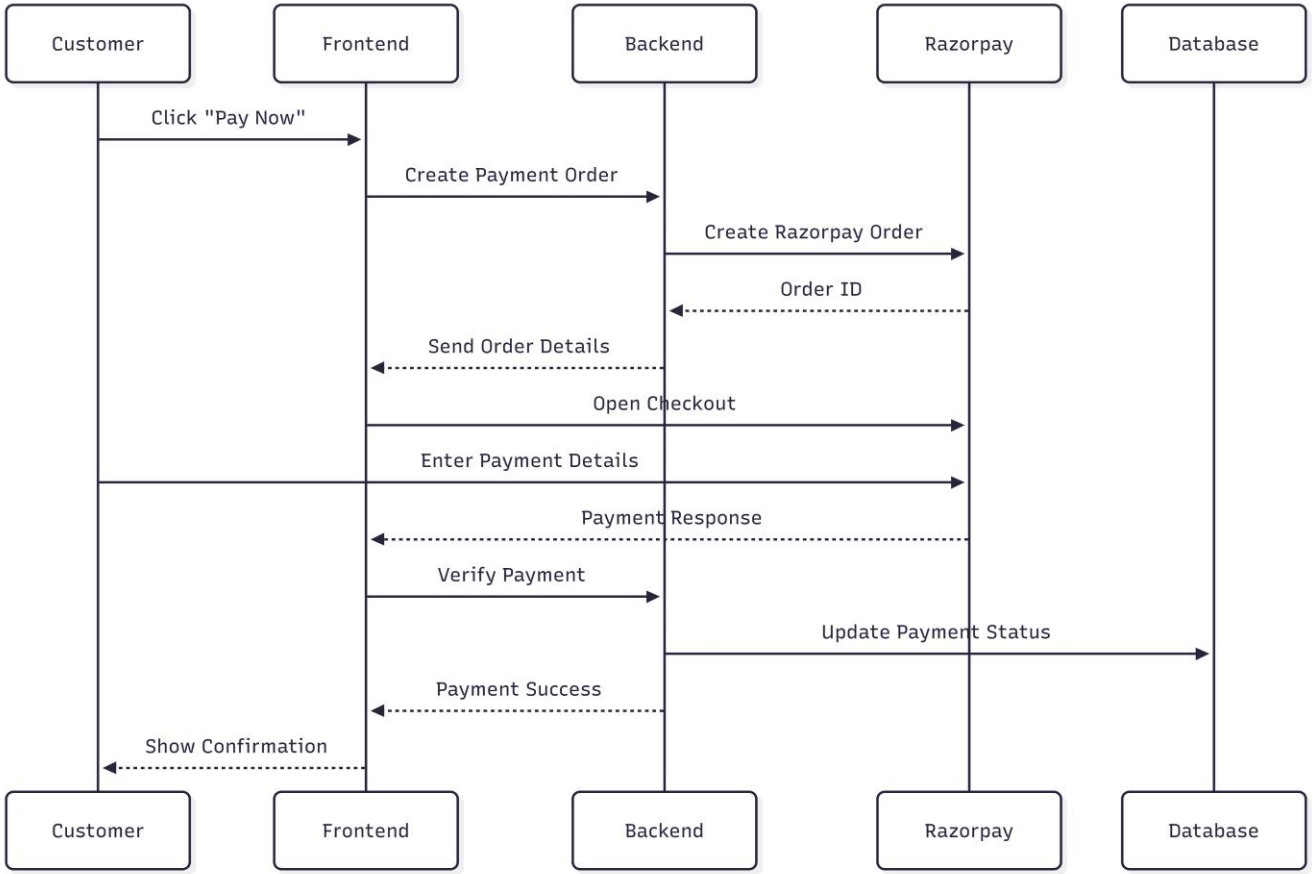


Figure 5.7 : Payment Integration Flow

6. Results and applications

Figure 6.1: Website Sitemap of Rental Management System

- **Home Page**

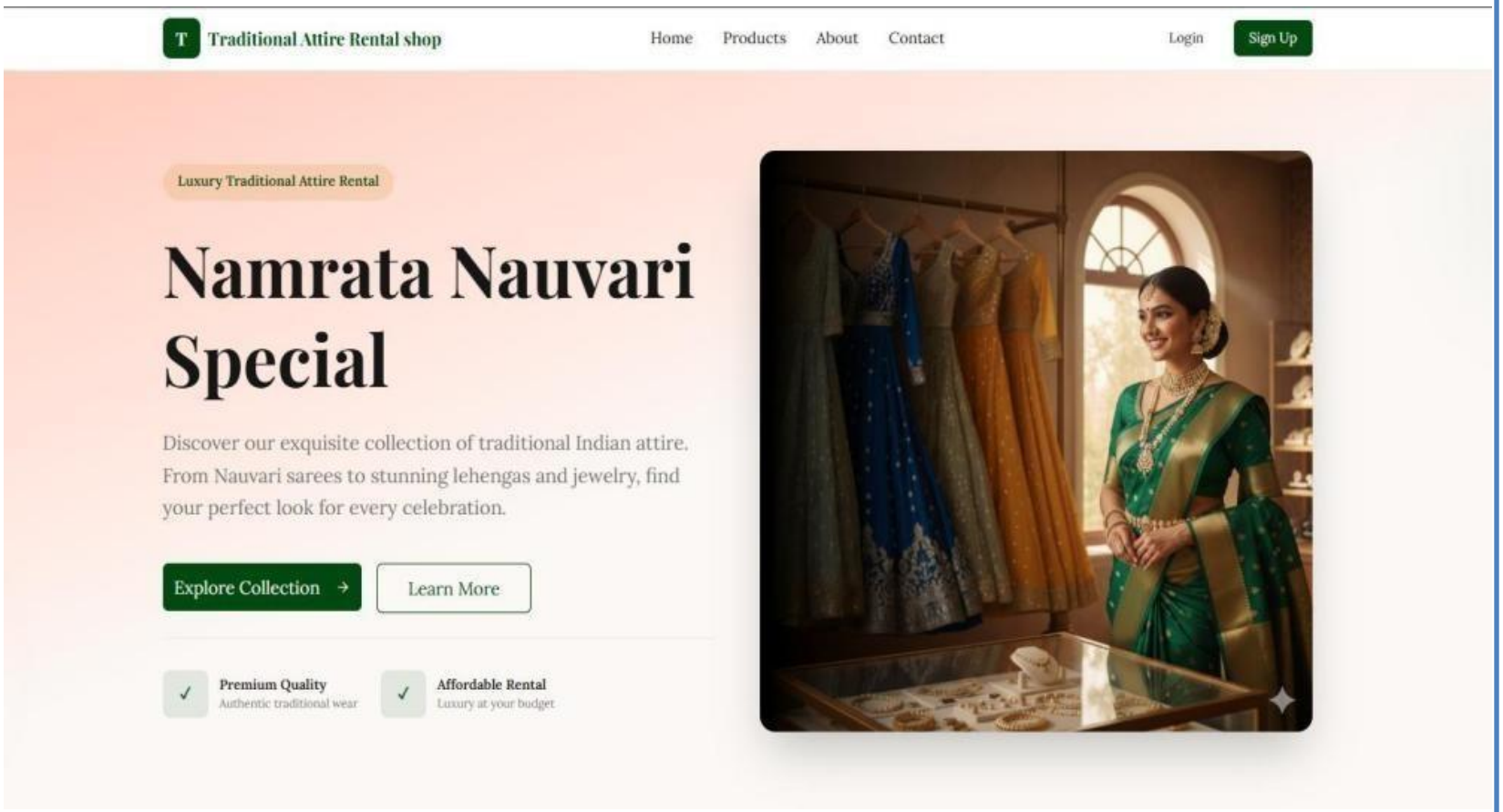
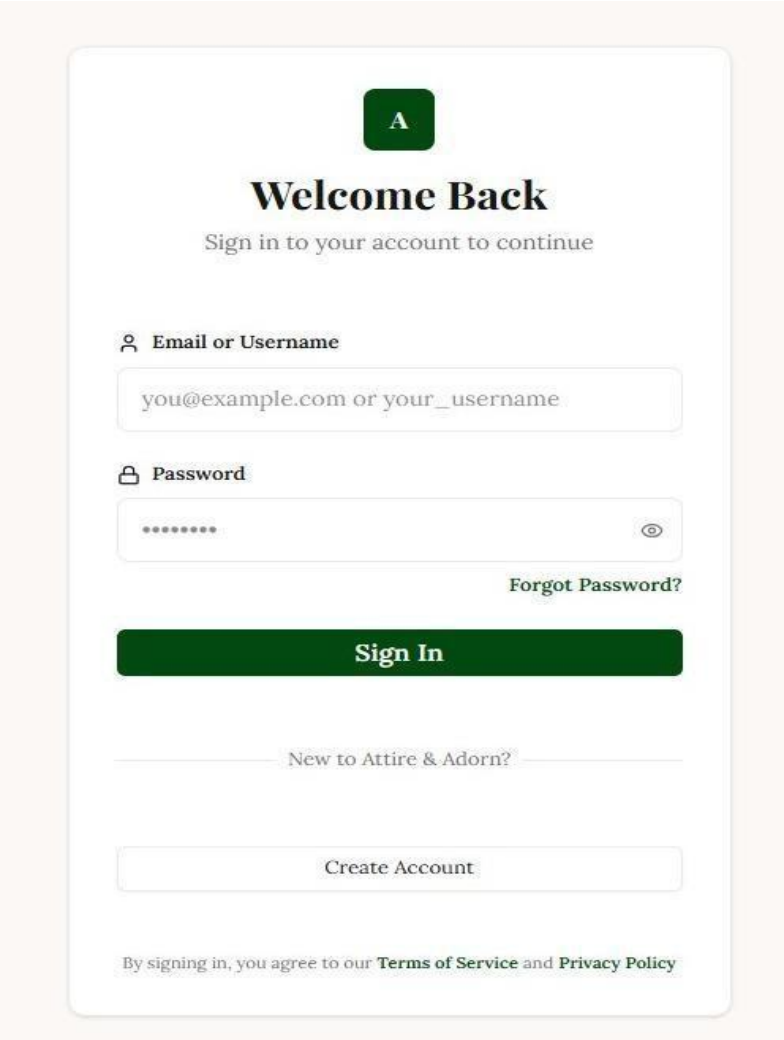


Figure 6.2: Home Page

- **Login Page**



- **Register Page**

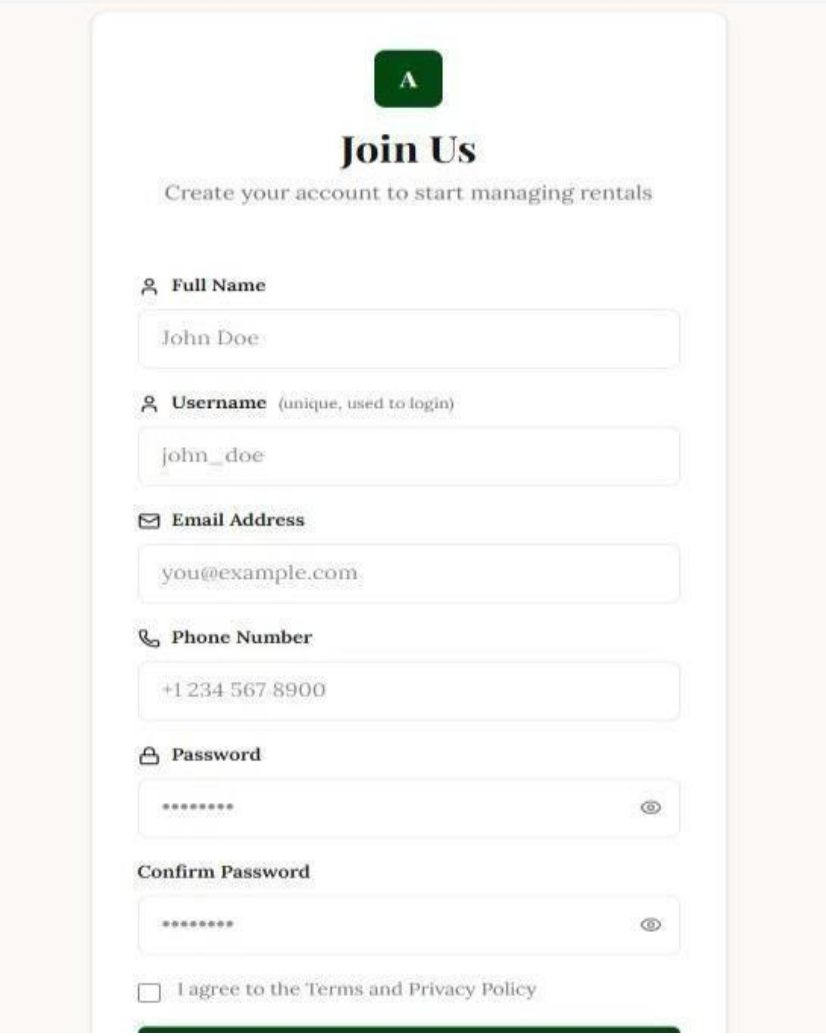


Figure 6.3: Login Page and Register Page

- **Product Page**

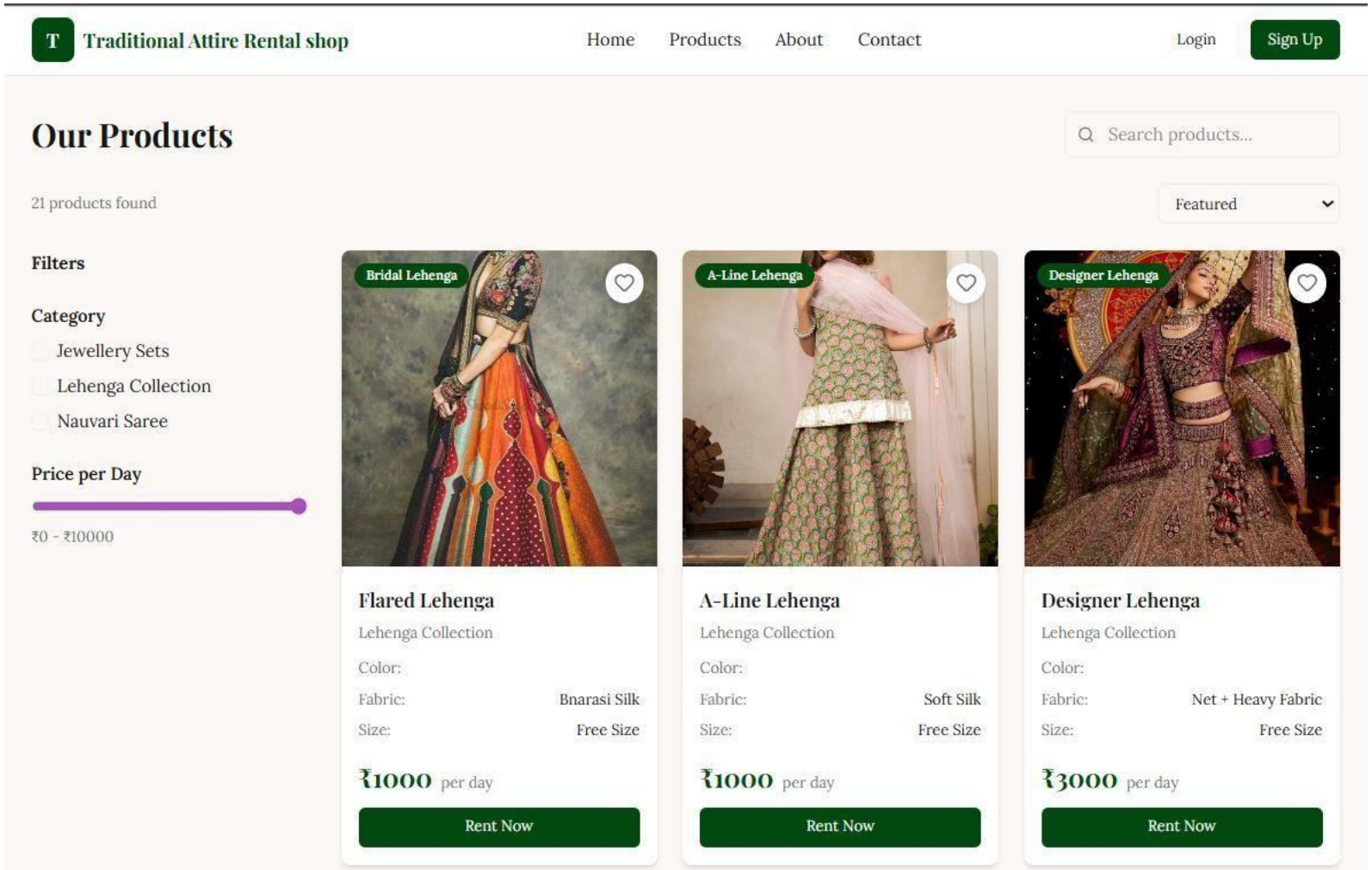


Figure 6.4: Products Page

- **Product Details Page**

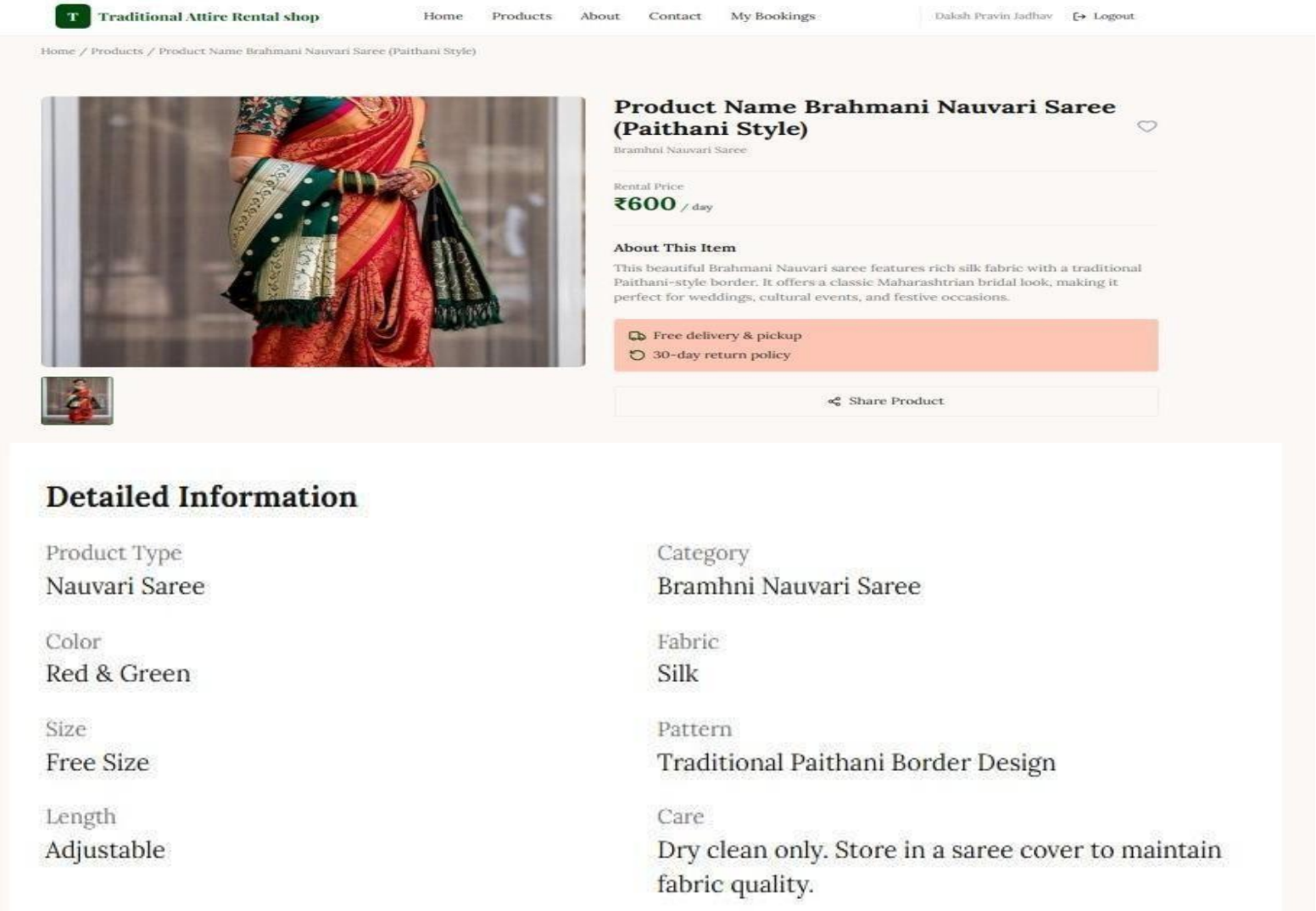


Figure 6.5: Product Details Page

- **Booking Form**

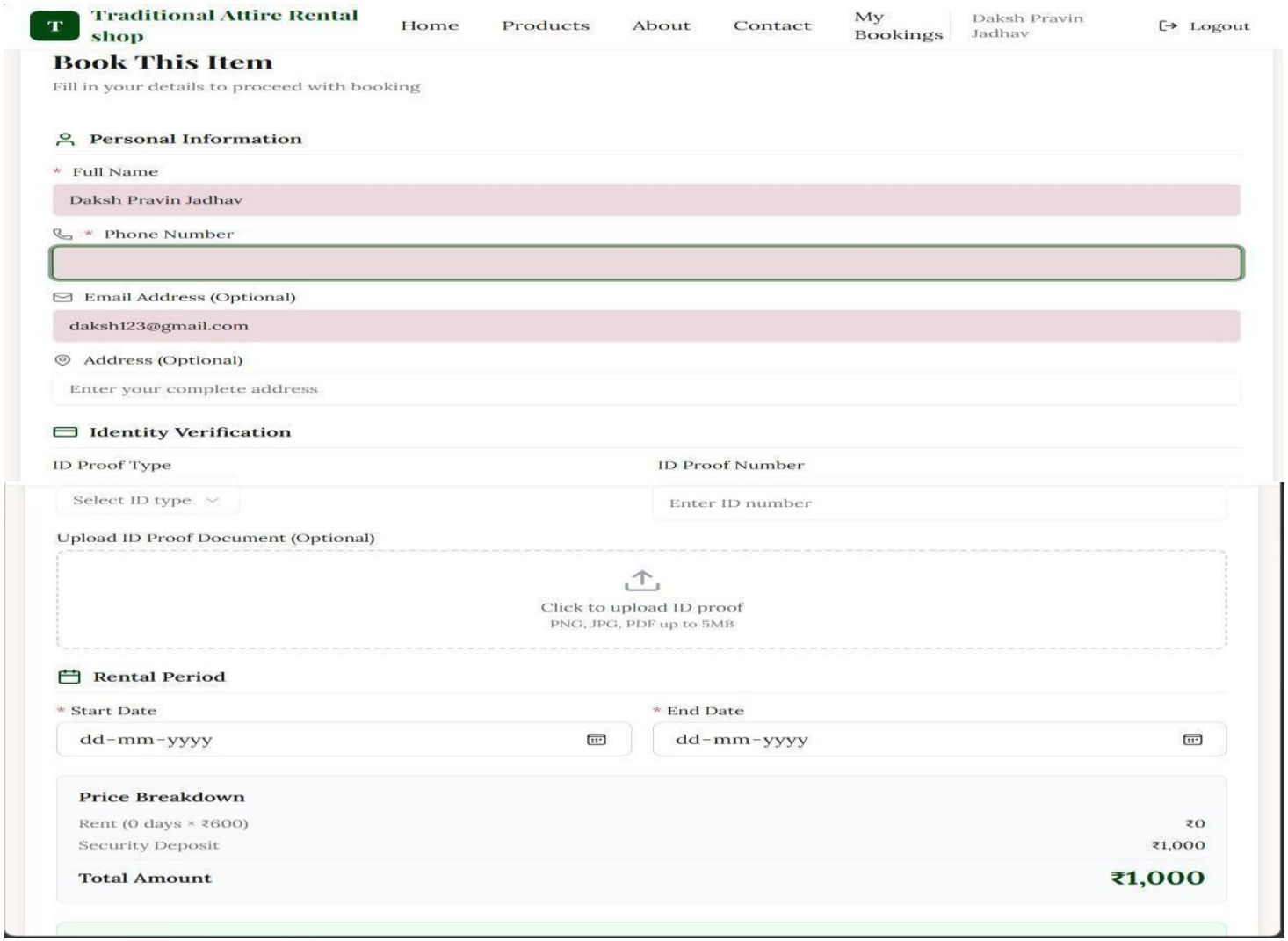


Figure 6.6: Booking Form

- **My Booking Page**

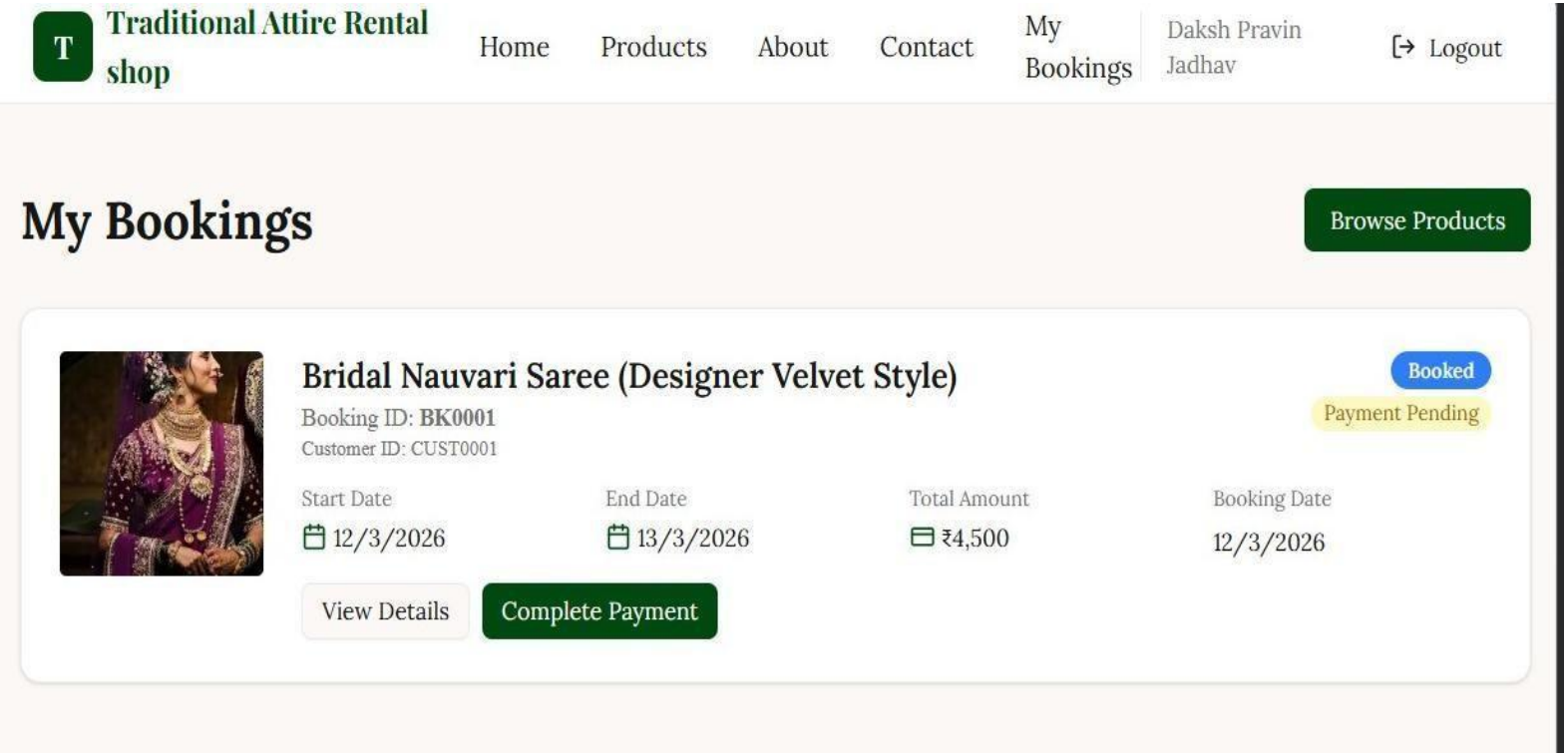


Figure 6.7: My Bookings Page

- **Booking Details Page**

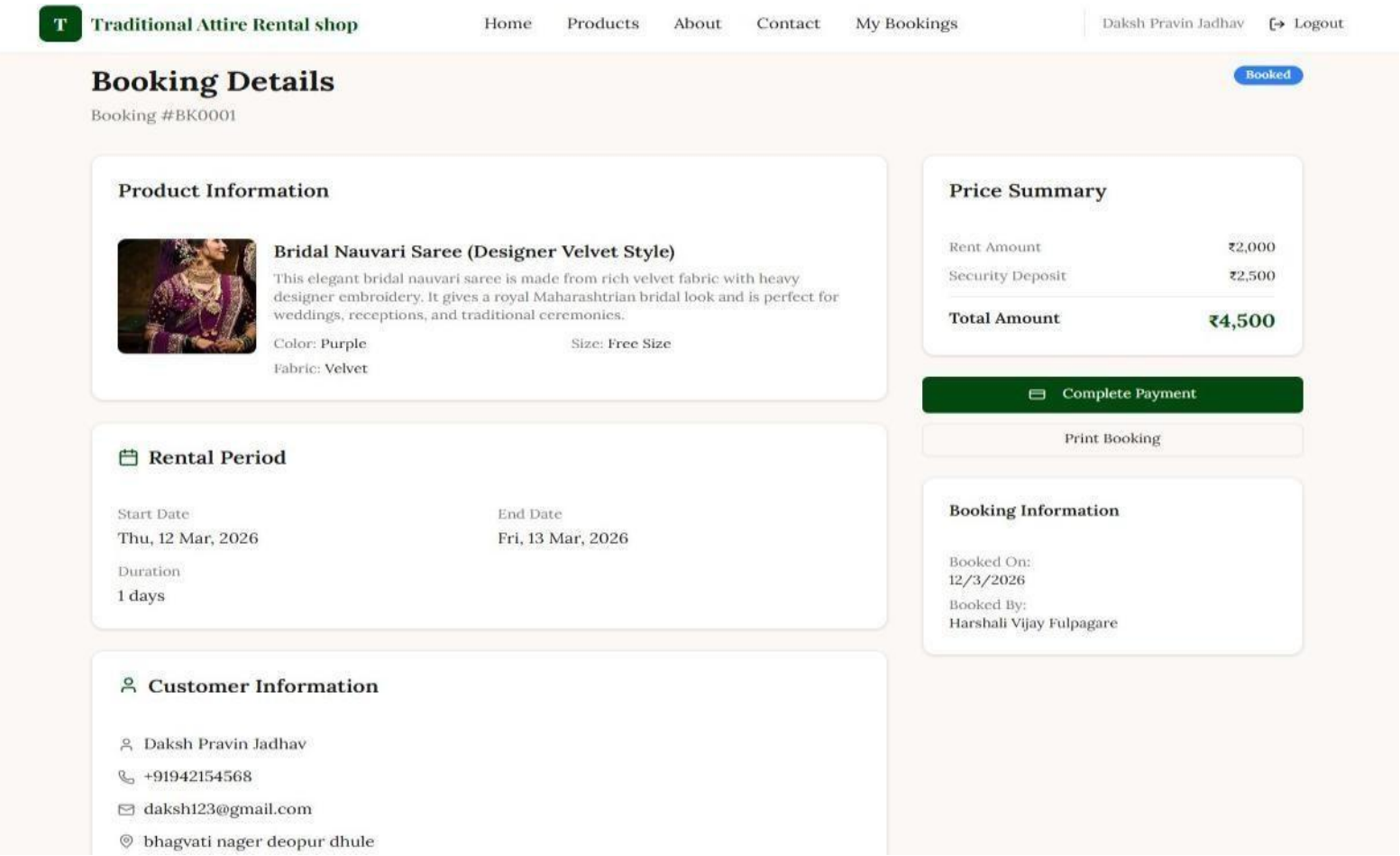


Figure 6.8: Booking Details Page

- **Booking Payment Page**

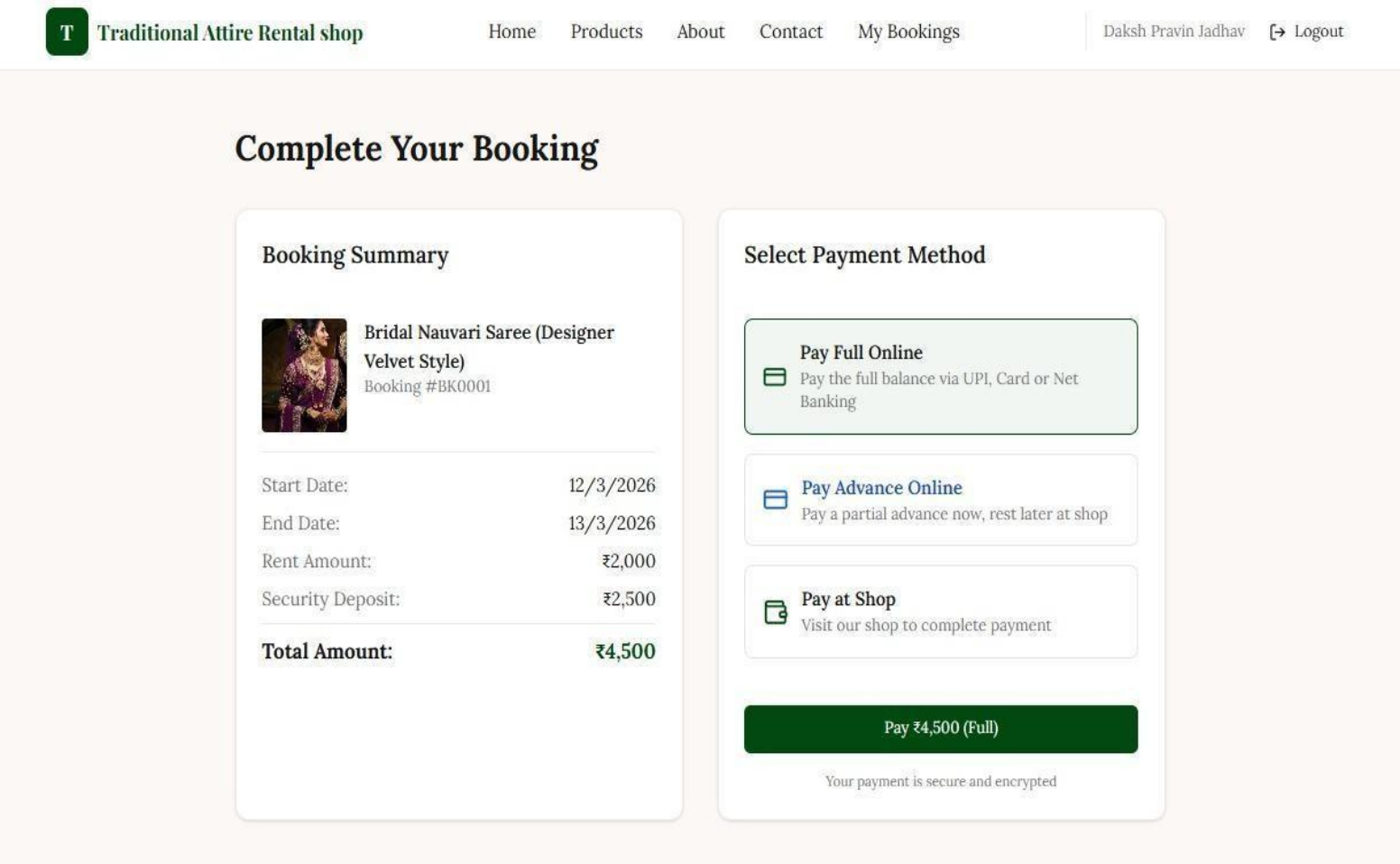


Fig. 6.9: Booking Payment Page

- **Admin Dashboard Page**

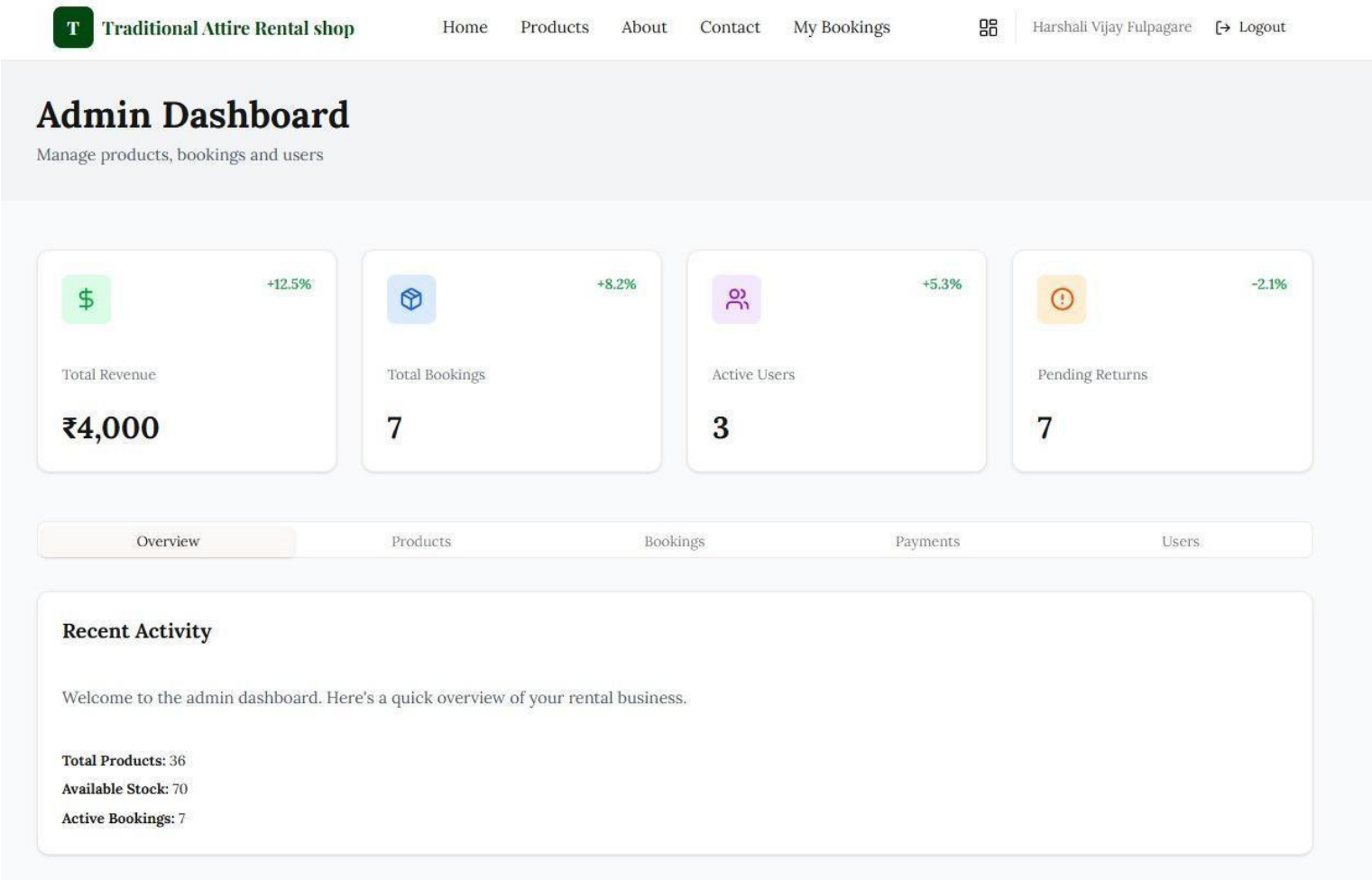


Figure 6.10: Admin Dashboard

- **Categories Management**

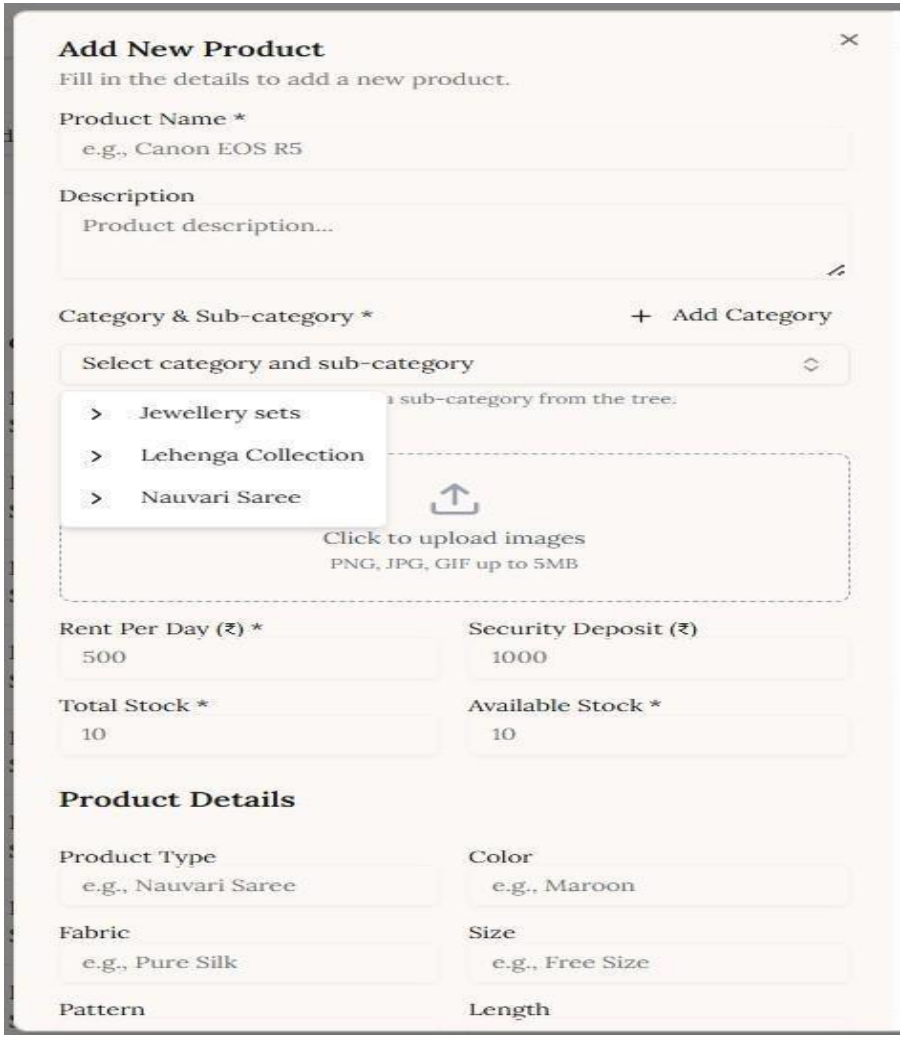


Figure 6.11: Categories Management

- **Product Management**

T Traditional Attire Rental shop Home Products About Contact My Bookings Harshali Vijay Fulpagare Logout

Product Management + Add Product

Code	Name	Category	Sub-category	Price/Day	Items	Stock	Status	Actions
PRD0016	Raj laxmi Nauvari Saree	Nauvari Saree	Raj Laxmi Nauvari Saree	₹700	PRD0016-1	1/1	Active	
PRD0015	Bridal Nauvari Saree	Nauvari Saree	Bridal Nauvari Saree	₹800	PRD0015-1 PRD0015-2	2/2	Active	
PRD0014	Jijau Nauvari saree	Nauvari Saree	Jijau Nauvari Saree	₹600	PRD0014-1	1/1	Active	
PRD0013	Shahi Mastani Nauvari Saree	Nauvari Saree	shahi mastani Nauvari Saree	₹800	PRD0013-1	1/1	Active	
PRD0012	Paithani Nauvari Saree	Nauvari Saree	Paithani Nauvari Saree	₹700	PRD0012-1	1/1	Active	
PRD0011	peshwai Nauvari Saree	Nauvari Saree	Peshwai Nauvari Saree	₹600	PRD0011-1	1/1	Active	
PRD0010	Lavani Nauvari Saree	Nauvari Saree	Lavani Style Nauvari Saree	₹600	PRD0010-1 PRD0010-2	2/2	Active	
PRD0009	Bramhni Nauvari Saree	Nauvari Saree	Brahmani Nauvari Saree	₹500	PRD0009-1	1/1	Active	
PRD0008	WaistBelt Jewllery	Jewellery sets	Waistbelt Jewellery	₹500	PRD0008-1 PRD0008-2 PRD0008-3	3/3	Active	
PRD0007	Oxidized Jewellery	Jewellery sets	Oxidised Jewellery	₹700	PRD0007-1 PRD0007-2	2/2	Active	
PRD0006	Maharashtrian Jewellery	Jewellery	Maharashtrian	₹1000	PRD0006-1	1/1	Active	

Figure 6.12: Product Management

- **Customer Management**

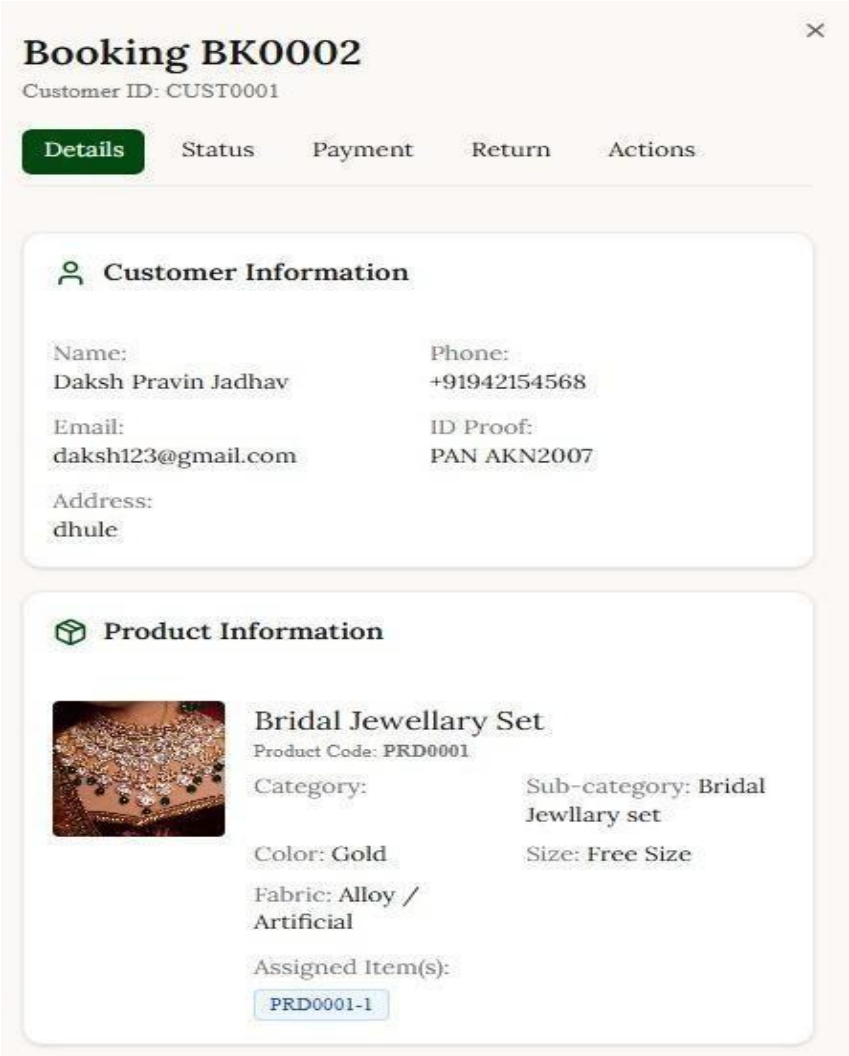


Figure 6.13: Customer Management

- **Booking Management**

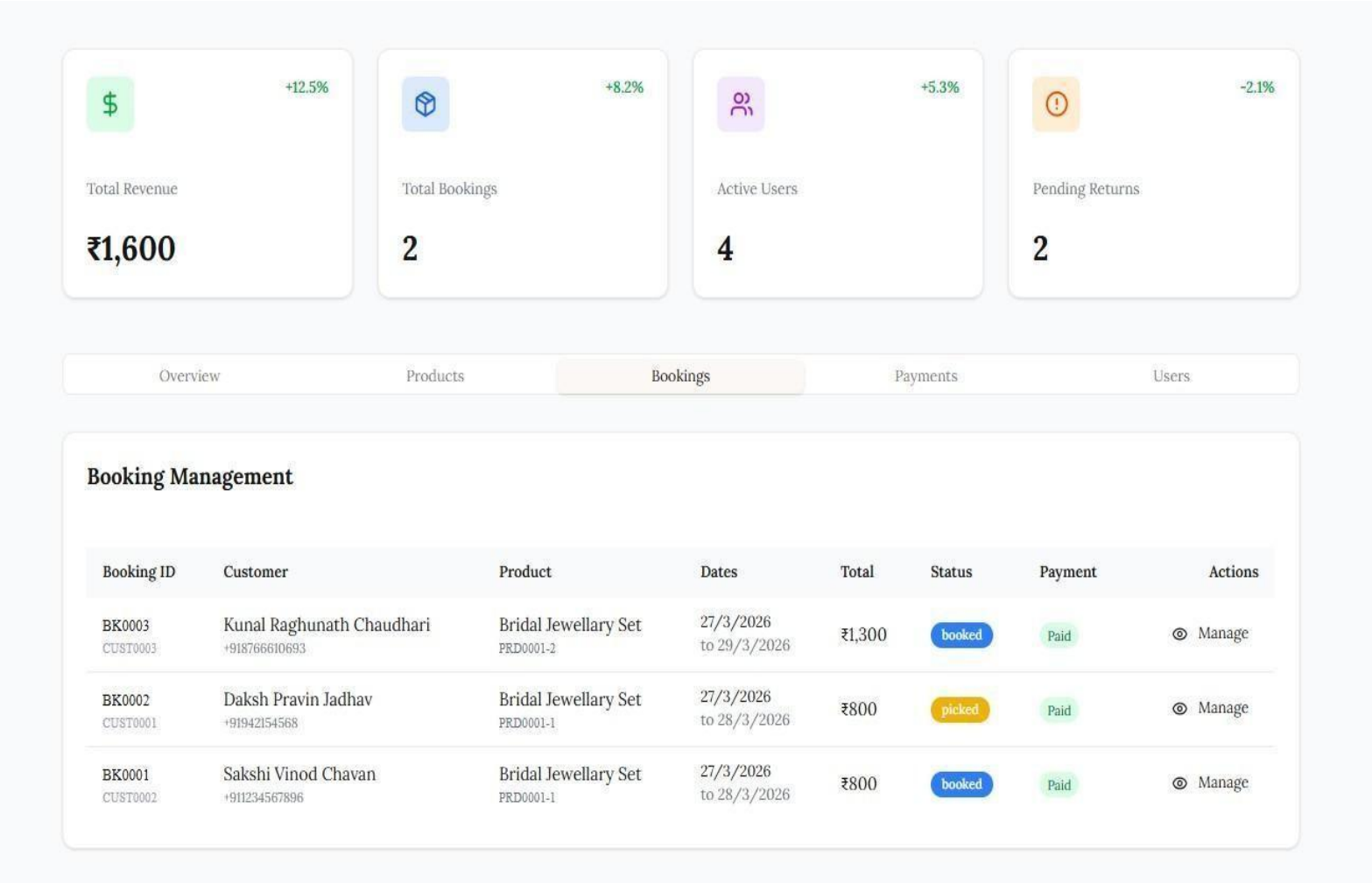


Figure 6.14: Booking Management

- **Inspection Management**

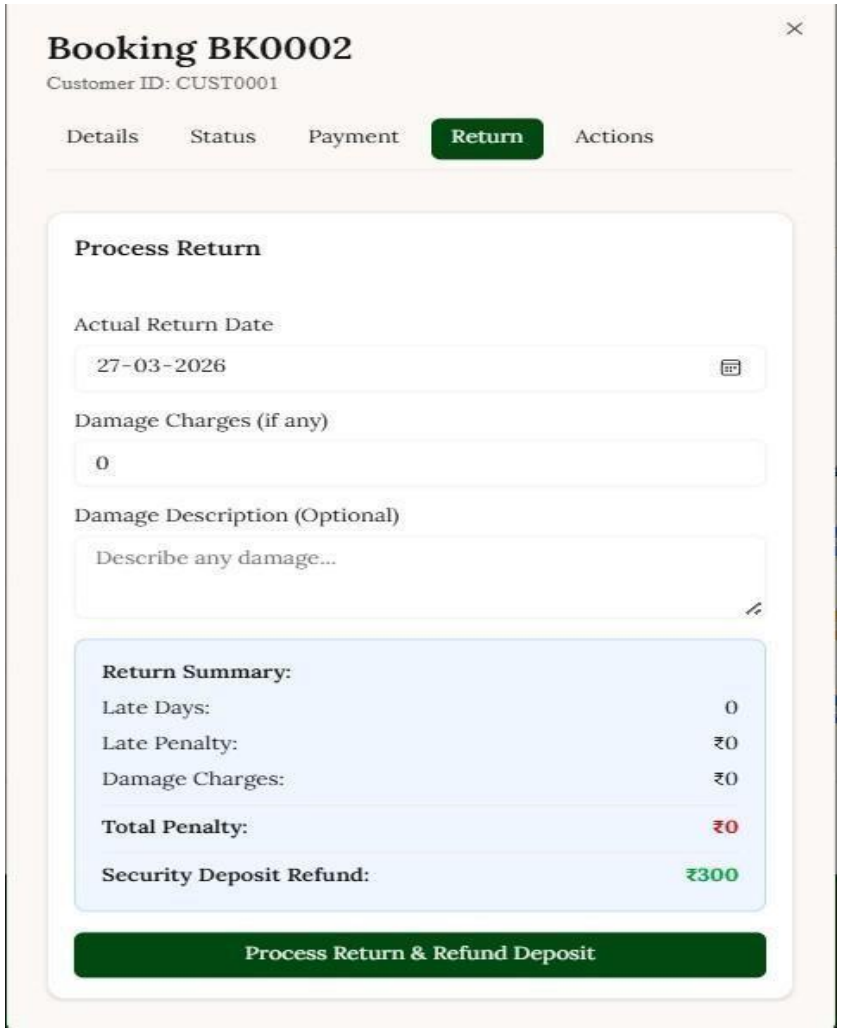


Figure 6.15: Inspection Management

- **Return Management**

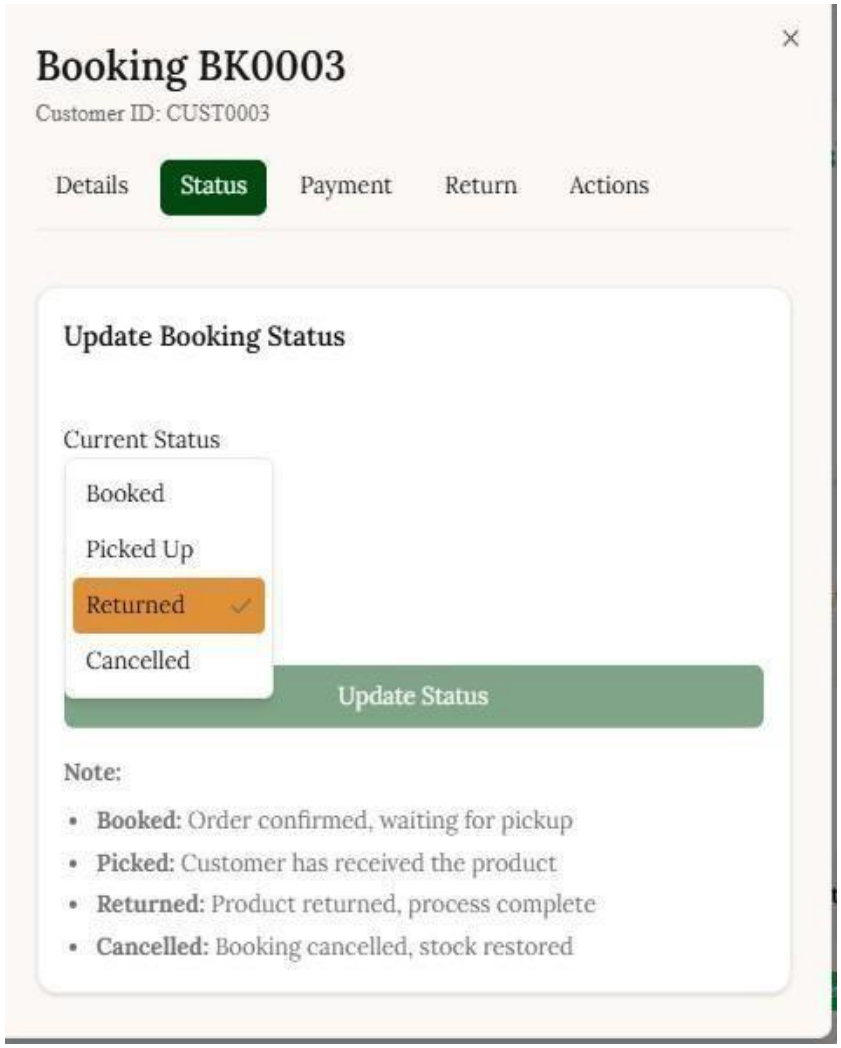


Figure 6.16: Return Management

- **Payment Management**

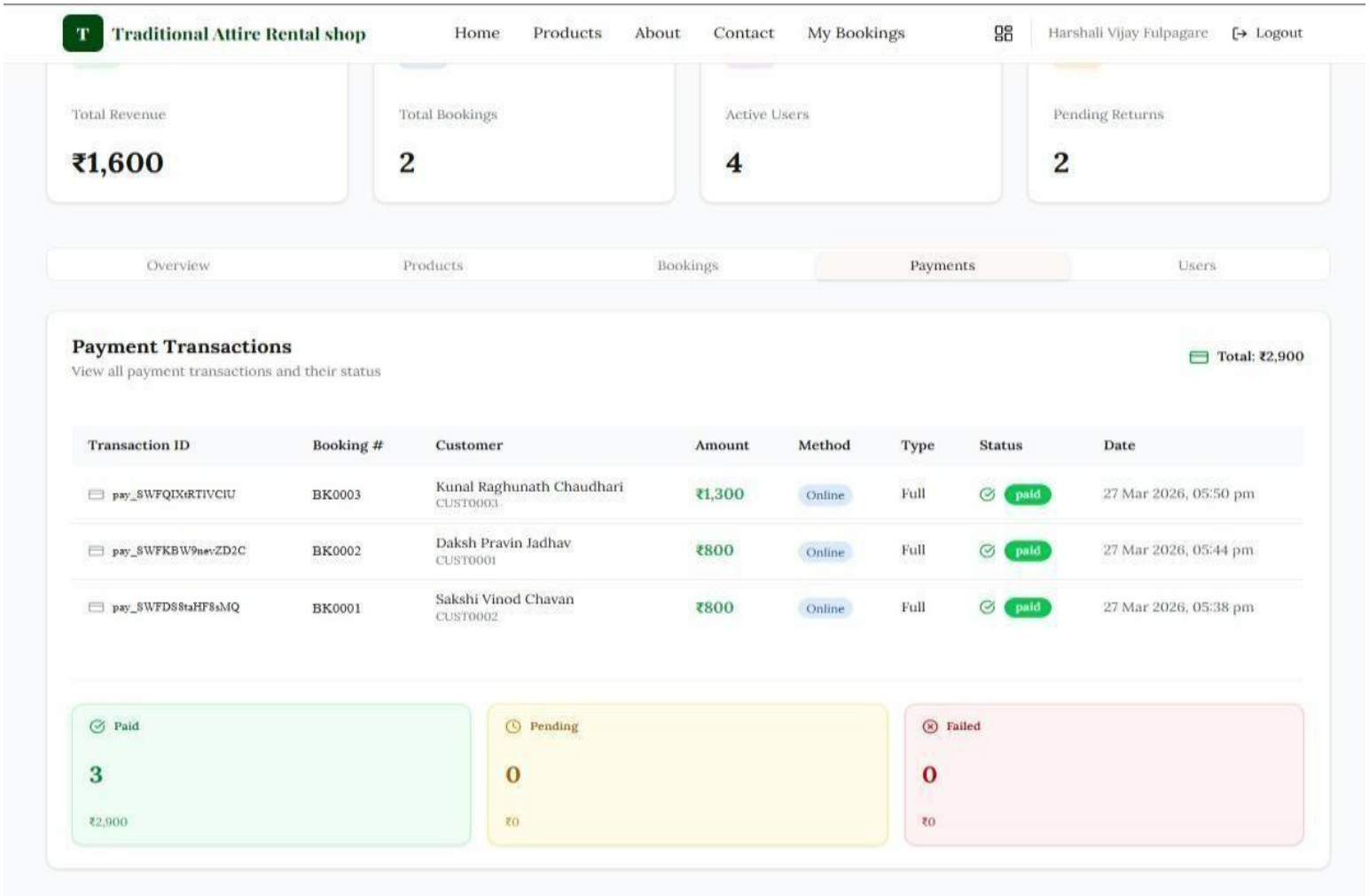


Figure 6.17: Payment Management

- **User Management**

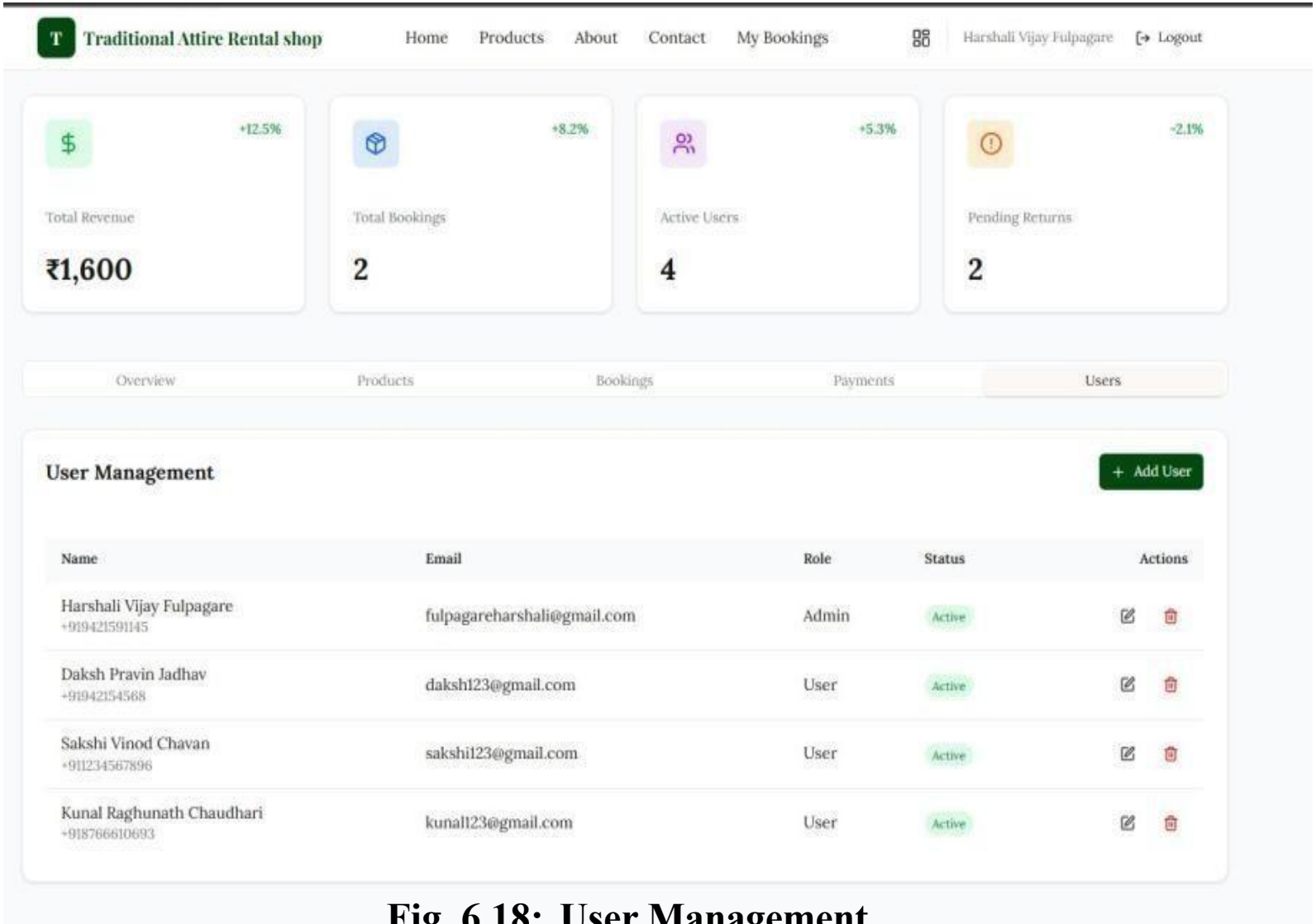


Fig. 6.18: User Management

6. Conclusion and future scope

7.1 Conclusion

The Traditional Attire Rental Management System successfully converts manual rental operations into an efficient digital platform. It integrates key features like product management, booking, payment, and reporting into a centralized system.

The system supports Admin, Staff, and Customer roles, ensuring smooth operations, real-time availability, and accurate record management. It reduces errors, minimizes booking conflicts, and improves overall efficiency and user experience.

Overall, the system provides a reliable and scalable solution for rental businesses, supporting better management and future growth.

7. Conclusion and future scope

7.2 Future Scope

- Instant notifications (SMS, WhatsApp, Email)
- Digital verification and e-signature
- QR/Barcode integration
- Mobile application (Android/Cross-platform)
- Online payment gateway integration
- Smart forecasting using data
- Advanced analytics dashboard
- Multi-branch support
- Audit and compliance logs
- Recommendation system for users

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